# When Do Local Income Taxes Cause Residential Sorting?

## Evidence from Pennsylvania's Piecewise EIT System

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**Abstract**

Do local income taxes cause residential sorting across sub-county boundaries? I use LEHD Origin-Destination data linking each worker's residence block to their workplace block at annual frequency, covering approximately 2,500 Pennsylvania municipalities and 500 school districts over 2002–2019.

Pennsylvania's Earned Income Tax operates a piecewise max-rule: each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where $\tau^{\text{home}}$ stacks the municipal and school-district resident rates and $\tau^{\text{nr,work}}$ is the work-location nonresident rate. A statutory rate change only moves the bill when the changed side of the max binds. I exploit this structure with a continuous stacked difference-in-differences design that decomposes each rate event by direction and binding-state transition, generating a within-design placebo: rate hikes whose post-event home stack stays below the nonresident rate (Inc-nobind) do not change the bill and should not move residents.

The preferred causal estimate is at the census block, where both rate-setting jurisdictions enter at their native geography. A 1 percentage-point municipal resident-rate

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increase in the binding cell reduces the log count of workers living in the block by $-0.21$ log points (approximately 19%), with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The Inc-nobind within-design placebo is statistically null on the same count outcome.

The municipality-level Inc-bind coefficient is larger ($-0.40$ log points on residents working in their own municipality) but pre-trended; I report it as policy-jurisdiction magnitude calibration rather than as the causal headline. School-district rate changes do not produce robust evidence of sorting and are treated as an aggregation diagnostic. Industry decomposition shows consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, consistent with a local-services demand spillover from departing residents rather than direct nonresident-tax incidence.

# 1 Introduction

The Tiebout hypothesis predicts that residents sort across jurisdictions which offer different fiscal amenities and different costs and residents will 'vote with their feet' and choose specific bundles of amenities for a given amount of taxation, called the Tiebout margin **tiebout1956**. A survey of the literature conducted by **agrawalLessonsStateLocal** highlights the current findings of state and local income taxation and identifies sub-state (county, municipality, school districts) income-tax sorting as one of the field's open questions. The literature has focused mostly on state tax adjustments on high-earners (**young2016**; **agrawalRelocationRichMigration**
The research shows that top-income households are generally mobile when rates between states are large, and **agrawal2015gradient** documents sharp discontinuities among commodity-taxes at sub-state borders, but does not estimate the sub-state Tiebout margin for income taxes. The closest existing research comes from **yangRESPONSIVENESSINCOMELOCAL2017** which finds small migration effects for county-level income tax adjustments in Indiana.

This paper estimates the municipal and school district level migration effects among workers in Pennsylvania from 2002–2019, by merging local income tax data with the LEHD Origin-Destination Employment Statistics (LODES), which links each worker's residence to their workplace at annual frequency and at sub-municipal geography. In Pennsylvania there are roughly 2,500 municipalities and 500 school districts (SD) that set their own Earned Income Tax (EIT) rates (**bagchiEffectsIntroducingWithholding2022**; **agrawalLessonsStateLocal**), producing a deep cross-section of multiple sub-state income-tax opportunities. The EIT offers two additional dimensions of variation that my empirical design exploits. First, the EIT is set independently by two overlapping rate-setting jurisdictions (municipalities and school districts) and a resident's home-side tax bill is the sum of the municipal resident rate and the school-district rate. This paper's primary focus is the municipal resident rate while the school district rate enters the empirical design but is treated as secondary. Second, the EIT operates using a piecewise maximization rule where each worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$, where $\tau^{\text{home}} = \tau^{\text{res,muni}} + \tau^{\text{SD}}$ is the sum of the municipal resident

rate and the school-district resident rate at the worker's address (the stack), and $\tau^{\text{nr,work}}$ is the nonresident rate at the work location. A statutory home-side rate change only moves the bill if the home stack is on the binding side of the max. Act 32 of 2008 (effective 2012) consolidated EIT collection into county-based Tax Collection Districts and mandated employer withholding at the higher of the two rates, which made the max-rule mechanically enforced. **bagchiEffectsIntroducingWithholding2022** estimates that Act 32 resulted in a 14% increase in compliance and much higher local revenues. My paper does not analyze compliance but rather estimates residential-sorting and provides additional analysis pre-post Act 32.

Beyond contributing to the general literature on local tax rate changes and residential sorting this paper's methodological contribution is utilizing the piecewise structure of the EIT to generate a binding-state decomposition to test how rate changes that move the tax bill from the non-resident side to resident side and producing a within-design placebo. A statutory rate change moves the worker's bill only when the home stack is on the binding side of the maximization function, but that same statutory change produces a predicted-null when the home stack stays below the work-side nonresident rate. Splitting rate-change events by whether or not they are binding isolates whose bill actually moves and those whose bill does not. Splitting rate-change events by binding state isolates the rate hikes that actually move the worker's bill and this within-design heterogeneity is what **agrawalLessonsStateLocal** identifies as missing from the existing U.S. local-income-tax literature. The decomposition is implemented at three units of observation (municipality, school district, census block) in a continuous stacked difference-in-differences design that uses never-changed municipalities as the control pool.

The max-rule turns a single statutory rate change into multiple heterogeneous effects depending on the binding-state transition. For each rate-setter side $s \in \{\text{muni}, \text{SD}\}$ I classify each rate event by direction (Increase or Decrease). Within that grouping I further decompose them by the binding-state transition of the home rate changes relative to the work-side

nonresident rate (bind, becomes binding, never binds) and six cells per rate-setter side: the three Inc cells (Inc-bind, Inc-becomes, Inc-nobind) and the three Dec mirrors. I am only able to identify causal effects for the increase cells because the decrease cells are lack enough observations (eleven Dec-bind, four Dec-becomes-non, fifteen Dec-nobind municipalities in total). The discussion below focuses on the three Inc cells, with decrease results reported alongside for completeness. The three cells differ in the implied change in the worker's bill $\Delta T = \max(\tau^{\text{home}}_{\text{post}}, \tau^{\text{nr}}) - \max(\tau^{\text{home}}_{\text{pre}}, \tau^{\text{nr}})$, which depends on which side of the max is binding pre and post; Section **??** walks through the institutional and arithmetic detail. In the Inc-bind cell the home stack was binding both pre and post, so the bill rises by the full $\Delta\tau$. In the Inc-becomes cell the home stack flips from non-binding to binding, so the bill rises only partially (the portion above the nonresident rate). The institutional rule's per-pp prediction is Inc-bind > Inc-becomes in magnitude because the per-pp bill change is larger, though a salience-based argument cuts the other way: Inc-becomes workers experience a discrete switch from paying zero home-side tax to paying a positive amount, which could trigger an outsized behavioral response on the threshold-crossing margin. The data favor the institutional prediction at both estimating units. In the Inc-nobind cell the home stack stays below the nonresident rate throughout, so the bill does not move and the predicted response is zero. Inc-nobind is therefore a within-design placebo: if the same statutory increase produces a response in Inc-bind and no response in Inc-nobind, the mechanism interpretation is credible; if Inc-nobind moves, the mechanism interpretation fails.

At the block level, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by $-0.21$ log points, approximately 19%, with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The Inc-nobind within-design placebo is statistically null on the same count outcome. The block-level decomposition is the preferred causal estimate because both rate-setting jurisdictions—the municipality and the school district—enter at their native geography and the pre-trend is clean. The municipality-level Inc-bind coefficient is larger in absolute value ($-0.40$ log points on the

count of residents who both live and work in the municipality, implying a $\approx 33\%$ decline per pp rate hike), and passes the joint Wald pre-trend test ($p = 0.23$). The block-level analogue is the preferred causal headline because the block observes both rate-setting jurisdictions at their native geography. I therefore report the municipality-level estimate as policy-jurisdiction magnitude calibration rather than as a second causal headline. School-district rate changes do not produce robust evidence of sorting and are treated as an aggregation diagnostic. Industry decomposition shows that consumer-facing jobs fall in the binding cell while tradeable jobs are insulated, consistent with a local-services demand spillover from departing residents rather than with direct nonresident-tax incidence on jobs.

The narrow, credible claim of the paper is that local income taxes cause residential sorting when the institutional rule makes them mechanically change the worker's bill—that is, in the Inc-bind cell, and at the municipality level also in the Inc-becomes cell where the hike flips the home stack from non-binding to binding ($\beta = -0.181^{**}$ on the count of residents who both live and work in the municipality). The pooled Inc/Dec specification that a conventional local-tax design would estimate is reported as a benchmark, but it mixes binding and non-binding events. The binding-state decomposition is the paper's contribution.

The remainder of the paper is organized as follows. Section **??** describes the institutional setting and Act 32. Section **??** introduces the six-cell binding-state decomposition. Section **??** describes LODES, the DCED rate panel, and the F-33 school-district controls. Section **??** sets out the continuous stacked difference-in-differences design at the three estimating units. Section **??** reports the pooled benchmark, the block-level causal headline, the municipality-level calibration, and the industry mechanism. Section **??** reports robustness diagnostics, and Section **??** concludes.

# 2 Institutional Background

Among U.S. states that authorize sub-state income taxation, Pennsylvania has the deepest cross-section of jurisdictions setting their own rate: approximately 2,500 municipalities and 500 school districts each levy a separate Earned Income Tax (EIT) on essentially all earned income earned in or by Pennsylvania residents (**bagchiEffectsIntroducingWithholding2022**; **agrawalLessonsStateLocal**). Unlike property taxes, which are paid by owners and capitalized into home values, the EIT falls directly on workers and varies across municipal boundaries within the same metropolitan area, creating sharp spatial discontinuities in tax burden that can influence where workers choose to live and work.

## 2.1 Pennsylvania Earned Income Tax

Pennsylvania's EIT is mandatory for all municipalities under the Local Tax Enabling Act of 1965 (**paAct511˙1965**). It applies to all earned income with no minimum threshold; retirees are exempt. The standard combined cap is 1%, typically split as 0.5% municipal and 0.5% school district. Home Rule municipalities may exceed this cap (**paHomeRule**), and Philadelphia operates separately under the Sterling Act of 1932 (**paSterlingAct˙1932**) at approximately 3.9%. For most non-Home-Rule boroughs and townships the EIT is the primary discretionary revenue lever, because property-tax millage limits and reassessment lags constrain the property side; quantifying that constraint would require a 2002–2019 municipality-level millage panel that is not currently available from DCED, and I leave the comparison to property-tax sorting as a separate exercise. Across 2002–2019, the panel contains 210 municipality resident-rate increases, 119 municipality decreases, 611 school-district increases, and 52 school-district decreases; the binding-state breakdown of these events is in Table **??**.

Each Pennsylvania municipality has three EIT rates relevant to a worker: the municipality resident rate $\tau^{\text{res,muni}}$ levied on residents of the municipality; the municipality nonresident

rate $\tau^{\text{nr,muni}}$ levied on workers employed in the municipality who live elsewhere; and the school-district rate $\tau^{\text{SD}}$ levied on residents only (school districts cannot tax nonresidents). Each address is associated with a six-digit Political Subdivision (PSD) code identifying its (municipality, school district) pair. Workers report this code on the Residency Certification Form, and employers use it to determine withholding obligations.

Under Act 32 (effective January 1, 2012), a worker's total EIT bill is determined by the maximum of the home-side rate stack and the work-location nonresident rate, and the bill is split piecewise between home and work jurisdictions:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr,muni}},\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr,muni}} \\ \left(0,\ \tau_i^{\text{nr,muni}}\right) & \text{if } \tau_j^{\text{res,muni}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr,muni}} \end{cases} \tag{1}$$

where $j$ is the worker's home municipality and $i$ is the work municipality. The work jurisdiction always collects its full nonresident rate first; the home municipality and school district then split whatever portion of the worker's bill is left over after the work claim. If the work nonresident rate exceeds the home stack, the home jurisdictions get nothing. The worker's total bill is the higher of the two stacks:

$$\text{Bill}_{ji} = \max\left(\tau_j^{\text{res,muni}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr,muni}}\right). \tag{2}$$

Equation (**??**) implies which margin a worker can use to respond to a rate increase. When the home stack binds ($\tau_j^{\text{home}} > \tau_i^{\text{nr,muni}}$), the worker can lower the bill only by moving home to a lower-rate municipality or school district. When the work nonresident rate binds, the worker can lower the bill only by changing job to a lower-nonres-rate work location; moving home without changing job has no effect. For workers who live and work in the same municipality, only the home stack applies—they pay $\tau_i^{\text{res,muni}} + \tau_i^{\text{SD}}$.

A worker who lives in municipality $j$ and moves to a lower-rate municipality $j'$ while keeping the same job at $i$ saves at least the SD rate at $j$: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$,

which is at most the old bill $\tau_j^{\text{home}}$ minus $\tau_j^{\text{SD}}$. Residence moves therefore do save tax under Pennsylvania's piecewise structure, because the home-side stack includes the SD rate, which the work jurisdiction cannot collect from nonresidents. This savings logic is built directly into the empirical design: the treatment $\Delta\tau^{\text{muni}}$ and the binding-state classification are constructed from the worker's home stack, so the regressions identify sorting in response to changes in the same quantity a relocating worker would optimize against.

Philadelphia is the one exception to the Act 32 framework. It operates under the Sterling Act of 1932: Philadelphia residents pay the city resident rate regardless of work location with no offset, and Philadelphia nonresidents pay the city nonresident rate, with no residual flowing to home jurisdictions in other Pennsylvania municipalities. My main estimates retain Philadelphia in the sample for completeness; results are robust to dropping it.

Act 32 of 2008 (**paAct32˙2008**) is central to identification in this paper. Before its 2012 effective date, EIT collection was radically decentralized: each of Pennsylvania's 2,500+ municipalities appointed its own tax officer. Nonresident EIT enforcement was weak because no centralized system linked employers, workers, and municipalities. A worker who moved out of a rate-raising municipality could, in practice, avoid paying the nonresident portion to the work municipality even though they legally owed it. The piecewise rule described in Equation **??** existed on paper but was not effectively enforced.

Act 32 transformed this system by consolidating collection into approximately 67 county-based Tax Collection Districts (TCDs), each with a single appointed collector. The law mandated standardized employer withholding at the higher of the employee's resident and work rate. This had two consequences. First, it made the EIT *mechanically binding*: rate changes now appear in every worker's paycheck through employer withholding, regardless of whether the worker resides in the taxing municipality. Second, it eliminated the possibility of *de facto* avoidance of the work-location nonresident claim through weak enforcement. After Act 32, a worker could no longer effectively duck the nonresident portion owed to the work municipality.

Importantly, Act 32 did not eliminate all tax savings from residential relocation. The piecewise rule in Equation (**??**) continues to imply that a worker whose original home stack binds can reduce the bill by moving to a sufficiently lower-stack home jurisdiction: the new bill is $\max(\tau_{j'}^{\text{home}}, \tau_i^{\text{nr}})$, which falls below the old home stack $\tau_j^{\text{home}}$ whenever $\tau_{j'}^{\text{home}} < \tau_j^{\text{home}}$ and the original home stack was the binding side. Act 32 thus changed enforcement and made the max-rule mechanically binding through withholding; it did not eliminate the residential-relocation margin that the mechanism in this paper exploits. **bagchiEffectsIntroducingWithholding20** documents the quantitative consequences of Act 32 directly: consolidated employer withholding increased PA EIT compliance, raised reported earned income in the rate-raising municipality, and shifted the de facto incidence toward workers. The present paper complements that work by isolating the residential-sorting response to the same withholding-enforcement regime, exploiting the cross-sectional variation in whether a given rate change actually moves the worker's bill (the binding-state decomposition); the pre/post-Act-32 split in Section **??** shows the binding-cell sorting response strengthens after Act 32, complementing the compliance-response finding.

School districts in Pennsylvania also levy an EIT under Act 511 (**paAct511˙1965**), typically at 0.5% (the other half of the standard 1% cap). Act 1 of 2006 (**paAct1˙2006**) (the Taxpayer Relief Act) additionally authorized school districts to exceed the 0.5% cap specifically to fund property tax relief. This generated a wave of school district rate increases in 2006–2011, creating rate variation that is largely independent of municipal fiscal decisions: the correlation between municipal and school district EIT rates across my panel is $-0.12$. Since every census tract sits in both a municipality and a school district, workers face two overlapping EIT rates that can change independently, providing a natural decomposition of the residential sorting response into municipal and school district components.

# 3 Conceptual Framework: Binding-State Decomposition

The piecewise tax allocation in Equation (**??**) is the analytical engine of the empirical design. A worker's bill is $T = \max(\tau^{\text{home}}, \tau^{\text{nr,work}})$. A rate change only *moves the bill* when its side of the max binds. This generates a sharp prediction: the same statutory rate change can produce a strong behavioral response, a moderate response, or no response at all depending on the binding state pre- and post-event.

## 3.1 Six Cells per Side

For each municipality $i$ and each rate variable $\tau \in \{\tau^{\text{res,muni}}, \tau^{\text{SD}}, \tau^{\text{nr,muni}}\}$, classify the municipality's first observed rate event by direction (Increase / Decrease) and binding-state transition relative to the municipality's own home-stack vs nonresident rate comparison: An *Inc-bind* event is a rate increase in which the home stack was binding both pre and post, so the bill rises by the full $\Delta\tau$ for in-municipality residents; this is the cell with the predicted strongest behavioral response. An *Inc-becomes* event is a rate increase that flips the home stack from non-binding to binding, so the bill rises by less than the full $\Delta\tau$ (only the portion above the work nonres rate), generating an intermediate predicted response. An *Inc-nobind* event is a rate increase in which the home stack stays below the nonres rate throughout, so the bill never moves; the predicted response is zero, and this cell serves as the within-design placebo.

The decrease cells mirror these definitions. A *Dec-bind* event is a rate decrease in which the home stack stays binding, so the bill falls by the full $\Delta\tau$—the predicted mirror of Inc-bind under symmetry. A *Dec-becomes-non* event is a rate decrease that flips the home stack from binding to non-binding, so the bill falls only down to the nonres rate. A *Dec-nobind* event is a rate decrease in which the home stack stays below the nonres rate throughout, so the bill never moves; this is the second placebo cell. A seventh category, NC (*never-changed*),

comprises municipalities whose $\tau$ never changes across the panel. NC serves as the control pool.

The fundamental testable prediction of the institutional rule is an ordered ranking of cell magnitudes: Inc-bind effects exceed Inc-becomes effects, which in turn exceed Inc-nobind effects, with the last close to zero in absolute value and serving as the within-design placebo. The central empirical comparison of the paper is between Inc-bind and Inc-nobind: if statutory rates matter mechanically through tax liability, the residential response should appear in Inc-bind and not in Inc-nobind. If Inc-nobind moves, the mechanism interpretation fails—the response is something other than the binding rule, because Inc-nobind events do not change the worker's bill. This makes the design self-falsifying in a way that a conventional pooled Inc/Dec specification is not: in the pooled specification, every increase is a treated event, while here the institutional rule predicts that only a subset of statutory increases should generate a response.

## 3.2 Three Treatment Margins

Pennsylvania's institutional structure separates three rate variables that change independently across municipality-years. The municipality resident rate $\tau^{\text{res,muni}}$ directly affects the bill of in-municipality residents through the home-stack channel. The school district rate $\tau^{\text{SD}}$ also affects the home stack, but is set by school districts rather than municipalities, so the predicted mechanism is the same binding-rule channel as the municipality rate even though the SD's institutional decision-making differs from the municipality's. The municipality nonresident rate $\tau^{\text{nr,muni}}$ affects nonresident workers commuting in, since the piecewise rule means it moves the bill of workers from low-home-stack origins.

Two predictions follow. First, each rate's binding-cell decomposition should show the same ranking of Inc-bind above Inc-becomes above Inc-nobind. Second, channel separation should hold: the municipality resident rate moves residents (live-and-work, total residents); the municipality nonresident rate moves nonresident workers (in-commuters); and the SD

rate's effect depends on whether SD changes induce similar residential responses.

# 4 Data

## 4.1 Data Sources

The analysis links two administrative systems that do not share common identifiers: LEHD/LODES residence–workplace records, keyed by Census block, and Pennsylvania EIT rate records, keyed by DCED municipality ID. The bridge is the TIGER 2020 block geometry, which nests blocks within county subdivisions (cousubs) by construction. A municipality-name crosswalk maps each DCED `municipality_id` to its corresponding COUSUBFP, matching 2,496 of 2,505 PA municipalities (99.6%). The remaining 9 are small boroughs that dissolved or merged during 2002–2019 and are dropped from the analysis. The full pipeline (scraper specs, PSD mapping, fiscal-control construction, auxiliary cross-state-inflow handling) is documented in the replication archive.

The residence–workplace data come from LEHD LODES Version 8, 2002–2019 (**lehdLODES**): block-level origin–destination links derived from state Unemployment Insurance wage records and the Quarterly Census of Employment and Wages (**abowdProducerDynamicsNew2009**). The Census Bureau applies synthetic noise infusion at the block level to protect respondent confidentiality, calibrated so that aggregation to tract and larger geographies approximates ground truth (**grahamDESIGNCOMPARISONLODES**). The sample period of 2002–2019 reflects LODES coverage beginning in 2002 and the need to drop 2020–2023, when COVID disrupted both commuting patterns and AFR fiscal data through ARPA transfers.

Municipal and school-district EIT rates come from the Pennsylvania Department of Community and Economic Development (DCED) MunStats portal (**padcedMunStats**), scraped annually. Pre-2012 nonresident rates, which DCED recorded only sporadically before Act 32 made employer withholding mandatory, are backfilled from the earliest observed post-2012 value per municipality. Two cleaning rules apply: DCED reports the flat-dollar Local Ser-

vices Tax in the same field as the EIT in some years, so rates above 5% are treated as data-entry errors and dropped (the highest standard PA EIT is ∼1%; only Philadelphia exceeds 3%), and rate values are aligned to April 1 of each year to match LODES's reference date.

Municipal fiscal controls come from the PA DCED Statewide Municipal Annual Financial Report database (**padcedAFR**). School-district fiscal controls come from the Census Annual Survey of School System Finances (F-33) (**censusF33**); the NCES `LEAID`-to-DCED `sd_id` crosswalk covers all 500 Pennsylvania districts. Two auxiliary sources support the pre-treatment balance and robustness exercises: FBI Uniform Crime Reports, Offenses Known and Clearances by Arrest, in the concatenated agency-year files compiled by **kaplanUCR**, and the Census Bureau Building Permits Survey (**censusBPS**), matched to PA municipalities at 93.5% and 93.3% rates respectively.

## 4.2 Descriptive Statistics

The primary outcome variable for the headline causal analysis is the block-level log count of workers living in the block. The count directly measures the residential location of active workers and avoids the denominator fragility of block-level shares, which are mechanically sensitive at sparse geography because both the numerator and denominator move. At the municipality and school-district aggregations I also report resident-worker counts, workplace counts, in-commuter flows, and share outcomes (*share_live_in*). Share outcomes at coarser geographies retain the interpretation of earlier literature; at the block level I treat them as supporting evidence rather than as the primary estimand. Fiscal outcomes used as falsification placebos and pre-treatment balance variables (log EIT revenue, log total tax revenue, log total revenue, log police expenditure, log population, log market value of real estate, plus crime and building permits) are reported in Appendix **??**.

This section reports summary statistics and key distributions of the estimation panels. Table **??** reports the municipality-year panel; Table **??** reports the block-year panel that

underlies the headline causal estimate. Table **??** reports the school-district aggregation. Table **??** summarizes the binding-state cell classification on the municipality resident-rate side, including the dose magnitude in each cell.

Table 1: Muni-year summary statistics (2002–2019)

| Variable | N | Mean | SD | p25 | p50 | p75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Resident rate $\tau^{M}$ (%) | 43,374 | 0.525 | 0.163 | 0.500 | 0.500 | 0.500 | 0.000 | 4.331 |
| Nonresident rate $\tau^{nr,M}$ (%) | 34,080 | 0.883 | 0.237 | 1.000 | 1.000 | 1.000 | 0.000 | 3.820 |
| SD rate $\tau^{SD}$ (%) | 43,374 | 0.622 | 0.289 | 0.500 | 0.500 | 0.500 | 0.000 | 2.050 |
| Home stack $\tau^{M} + \tau^{SD}$ (%) | 43,374 | 1.147 | 0.316 | 1.000 | 1.000 | 1.000 | 0.000 | 4.331 |
| log live-and-work | 40,748 | 3.638 | 2.011 | 2.079 | 3.714 | 5.088 | 0.000 | 12.877 |
| log total jobs | 45,291 | 5.819 | 2.006 | 4.489 | 5.883 | 7.234 | 0.000 | 13.353 |
| Share live-and-work in muni | 45,291 | 0.095 | 0.081 | 0.037 | 0.080 | 0.133 | 0.000 | 1.000 |
| Total jobs | 45,291 | 2076.691 | 13031.730 | 89.000 | 359.000 | 1385.500 | 1.000 | 629698.000 |

*Notes:* Muni-year level; one observation per (municipality, year). Rates in percentage points. $\tau^{nr,M}$ has reporting gaps before Act 32; the $N$ for that row is smaller than the full panel.

The municipality-year panel covers 43,374 observations. The 25th, 50th, and 75th percentiles of the resident rate are all 0.5%, reflecting the statutory cap that the vast majority of municipalities adopt; the right tail reaches 4.33% in Home-Rule and Act 47 cities. Nonresident rates have a smaller $N$ (34,080) because DCED's pre-Act-32 coverage of the nonresident-rate field is incomplete. The typical Pennsylvania municipality has ∼360 jobs (median) and a ∼8% share of workers who both live and work in the same municipality.

Table 2: Block-year summary statistics (2002–2019)

| Variable | N | Mean | SD | p25 | p50 | p75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| Home count (workers living) | 5,030,958 | 18.822 | 31.677 | 4.000 | 10.000 | 22.000 | 0.000 | 1572.000 |
| log home count | 4,795,213 | 2.293 | 1.198 | 1.386 | 2.398 | 3.135 | 0.000 | 7.360 |
| Work count (jobs in block) | 5,030,958 | 18.826 | 138.957 | 0.000 | 0.000 | 4.000 | 0.000 | 20945.000 |
| log work count | 1,894,432 | 2.256 | 1.663 | 1.099 | 2.079 | 3.367 | 0.000 | 9.950 |
| Share live-and-work in muni | 4,795,213 | 0.154 | 0.222 | 0.000 | 0.067 | 0.203 | 0.000 | 1.000 |
| Share work outside muni | 4,795,213 | 0.846 | 0.222 | 0.797 | 0.933 | 1.000 | 0.000 | 1.000 |
| Resident rate $\tau^{M}$ (%) | 4,782,928 | 0.762 | 0.799 | 0.500 | 0.500 | 0.500 | 0.000 | 4.331 |
| SD rate $\tau^{SD}$ (%) | 4,782,928 | 0.607 | 0.376 | 0.500 | 0.500 | 0.500 | 0.000 | 2.050 |
| Nonresident rate $\tau^{nr,M}$ (%) | 4,181,214 | 1.063 | 0.657 | 1.000 | 1.000 | 1.000 | 0.000 | 3.820 |
| Blocks per muni | 2,561 | 120.922 | 384.585 | 41.000 | 73.000 | 132.000 | 1.000 | 16550.000 |

*Notes:* Block-year level; one observation per (census block, year). Panel covers 309,680 blocks in 2,561 municipalities.

The block-year panel is the largest of the three (∼5.0M observations across ∼280,000 PA Census blocks). The median block has 10 workers living in it and zero workers working in it (work-side counts are right-skewed and concentrated in commercial blocks), so block-level share outcomes are mechanically fragile at sparse geography. The log(home count) regression sample restricts to blocks with at least one resident worker; $N = 4.8$M.

Table 3: School-district-year summary statistics (2002–2019)

| Variable | N | Mean | SD | p25 | p50 | p75 | Min | Max |
|---|---|---|---|---|---|---|---|---|
| SD rate $\tau^{SD}$ (%) | 49,226 | 0.610 | 0.278 | 0.500 | 0.500 | 0.500 | 0.000 | 2.050 |
| Resident rate $\tau^{M}$ (%) (muni-year) | 49,226 | 0.523 | 0.162 | 0.500 | 0.500 | 0.500 | 0.000 | 4.564 |
| Nonres rate $\tau^{nr,M}$ (%) (muni-year) | 38,536 | 0.881 | 0.237 | 1.000 | 1.000 | 1.000 | 0.000 | 3.967 |
| share_state_rev_sd | 9,278 | 0.416 | 0.166 | 0.279 | 0.414 | 0.555 | 0.000 | 0.821 |
| share_local_rev_sd | 9,278 | 0.536 | 0.186 | 0.376 | 0.536 | 0.691 | 0.103 | 1.000 |
| share_federal_rev_sd | 9,278 | 0.048 | 0.033 | 0.024 | 0.040 | 0.062 | 0.000 | 0.247 |
| share_property_tax_sd | 9,278 | 0.420 | 0.176 | 0.270 | 0.407 | 0.562 | 0.000 | 0.853 |
| share_instructional_sd | 9,278 | 0.522 | 0.070 | 0.492 | 0.536 | 0.567 | 0.000 | 1.000 |
| share_support_sd | 9,278 | 0.279 | 0.039 | 0.259 | 0.284 | 0.305 | 0.000 | 0.438 |

*Notes:* SD-year level. F-33 share controls are constructed from Census F-33 revenue/expenditure totals.

The SD-year panel is much smaller: 500 Pennsylvania school districts × 18 years, 9,278 observations with Census F-33 financial controls. School districts derive roughly 42% of revenue from state aid, 54% from local sources (of which property tax is the dominant component at 42% of total revenue), and 5% from federal sources. Instructional spending averages 52% of expenditure.

Table 4: Cell counts: muni resident-rate side, 2002–2019

| Cell | Munis | Muni-years | Mean dose (pp) | Max dose (pp) |
|---|---|---|---|---|
| NA | 1 | 18 | 0.000 | 0.000 |
| NC | 1,793 | 32,266 | 0.000 | 0.000 |
| Inc-bind | 40 | 720 | 0.191 | 1.500 |
| Inc-becomes | 58 | 1,044 | 0.339 | 4.331 |
| Inc-nobind | 26 | 468 | 0.273 | 0.975 |
| Dec-bind | 11 | 198 | -0.104 | 0.350 |
| Dec-becomes-non | 4 | 72 | -0.217 | 0.000 |
| Dec-nobind | 15 | 270 | -0.283 | 0.700 |

*Notes:* Cells classify each muni's first resident-rate event by direction (Inc/Dec) and the binding-state transition relative to the muni's nonresident rate. NC is the never-changed control pool. Dose is the cumulative $\Delta\tau$ at event-time $k = +5$.

The binding-state decomposition assigns the ∼2,500 municipalities into seven cells based on their first municipality-rate event. 1,793 municipalities are NC controls and never change their resident rate; 40 are Inc-bind (the headline cell), 58 are Inc-becomes, 26 are Inc-nobind (within-design placebo), and 30 are Dec cells combined. The mean cumulative Inc-bind dose at event-time $k = +5$ is 0.19 pp (Inc-becomes 0.34 pp, Inc-nobind 0.27 pp); the upper tail reaches 1.5 pp for Act 47 cities.

Figure **??** plots the empirical CDF of the three rate variables that enter the piecewise max-rule: the municipality resident rate $\tau^{M}$, the municipality nonresident rate $\tau^{nr,M}$, and the home stack $\tau^{M} + \tau^{SD}$. Under Act 511 of 1965 (**paAct511˙1965**), the combined home-side EIT is capped at 1% for ordinary boroughs and townships and is conventionally split 0.5% municipal plus 0.5% school district; the vast majority of Pennsylvania municipalities sit at this statutory default and never adjust. Approximately 121 of the ∼2,500 municipalities change their resident rate at any point in the sample, and most school districts likewise hold at 0.5%. The ECDF makes this concentration visible: roughly 95% of municipality-years sit

at or below the 0.5% resident-rate cap, and the home-stack CDF reaches $\sim 78\%$ by the 1% combined cap. The long right tails correspond to two policy carve-outs: Act 47 distressed cities (**paAct47˙1987**) (Pittsburgh-area municipalities, Wilkes-Barre, Hazleton, Johnstown, and others) that exceeded the municipality half of the 1% cap, and Act 1 school districts (**paAct1˙2006**) that converted property-tax authority into EIT authority. Philadelphia operates under the Sterling Act of 1932 (**paSterlingAct˙1932**) rather than Act 511 and accordingly sits at the far right of the resident-rate distribution. These tail municipalities and school districts generate the identifying variation that the binding-state decomposition then splits by whether each statutory rate change moved the worker's bill.

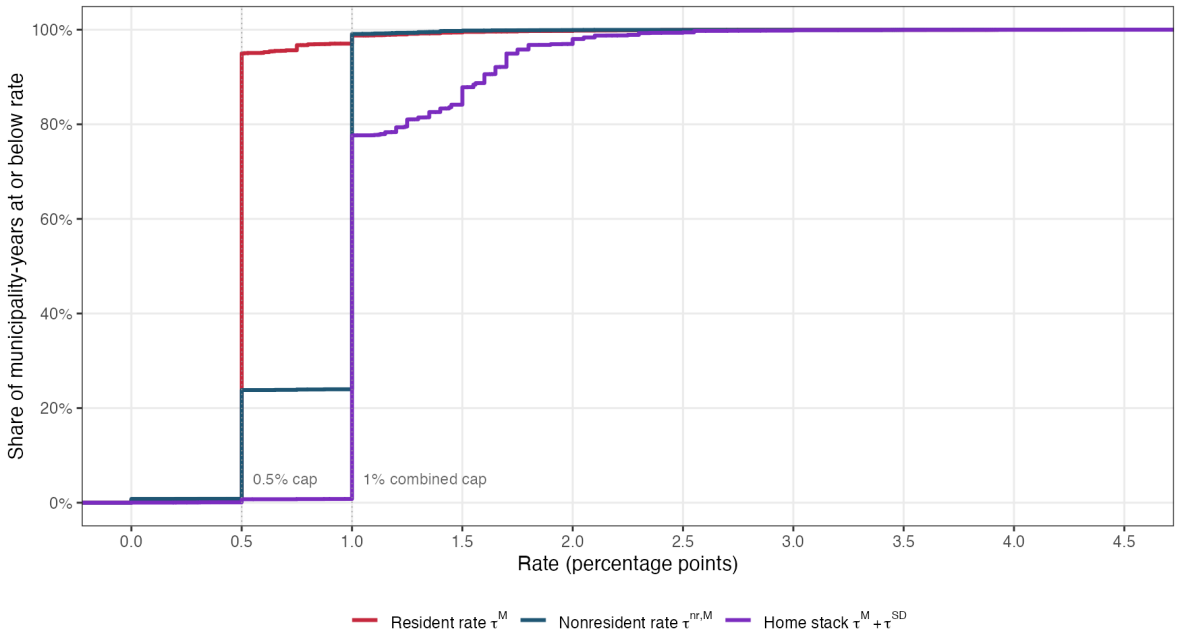



Figure 1: Empirical CDF of Pennsylvania EIT rates, 2002–2019. Each step shows the share of municipality-years at or below a given rate, separately for the municipality resident rate $\tau^{M}$ (red), the municipality nonresident rate $\tau^{nr,M}$ (blue), and the home stack $\tau^{M} + \tau^{SD}$ (purple). Dotted reference lines mark the 0.5% municipality cap and the 1% combined home-side cap under Act 511 of 1965. Approximately 95% of municipality-years sit at the 0.5% resident-rate cap and $\sim 78\%$ at the 1% combined home-stack cap; the long right tails are Act 47 distressed cities, Home-Rule cities, and Act 1 school districts.

Figure **??** reports the timing of municipal resident-rate changes across the panel. Increases outnumber decreases nearly two-to-one overall (210 increases vs. 119 decreases over

2002–2019), and the gap widens further after Act 32 took effect in 2012: roughly 80% of post-2012 rate-change events are increases, consistent with consolidated withholding raising the political cost of cutting a rate that is now mechanically enforced.

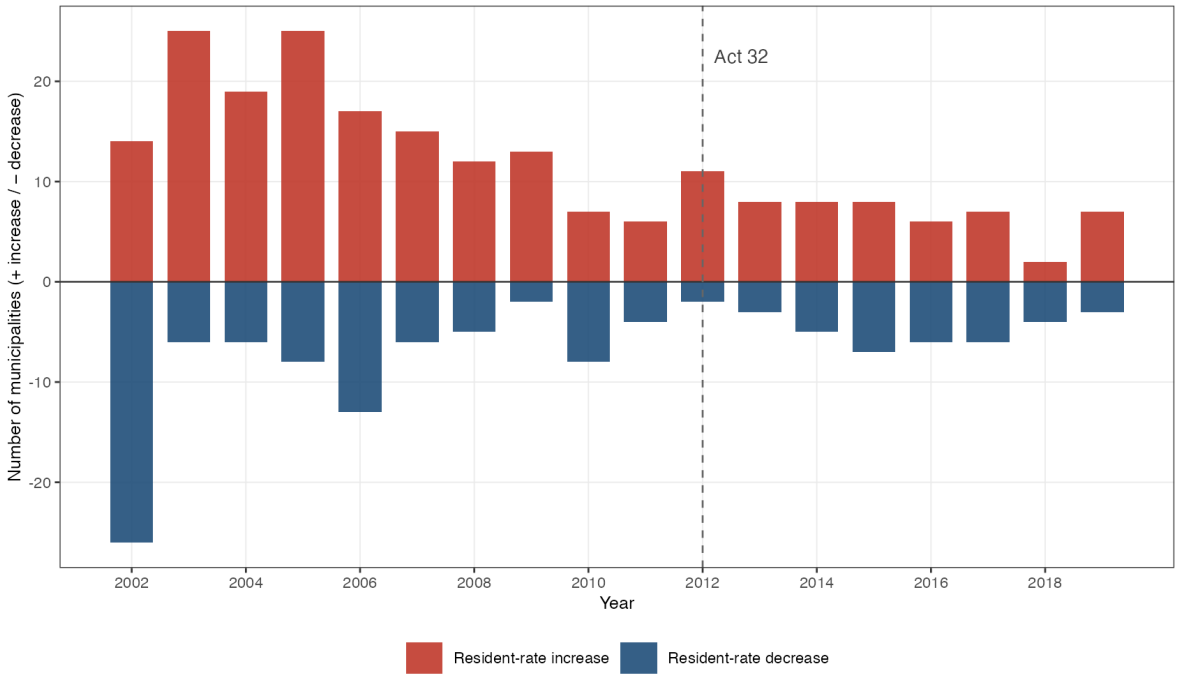



Figure 2: Number of PA municipalities changing their resident rate, by year. Increases shown above zero; decreases shown below. Dashed line: 2012, when Act 32 consolidated withholding.

Figure **??** documents the institutional break at Act 32. Before 2012, fewer than 60% of PA municipalities reported a nonresident rate to DCED; after 2012, when consolidated withholding made the max-rule mechanically binding, coverage rises sharply. This is the pre-/post-Act-32 enforcement break used in the heterogeneity test of Section **??**.

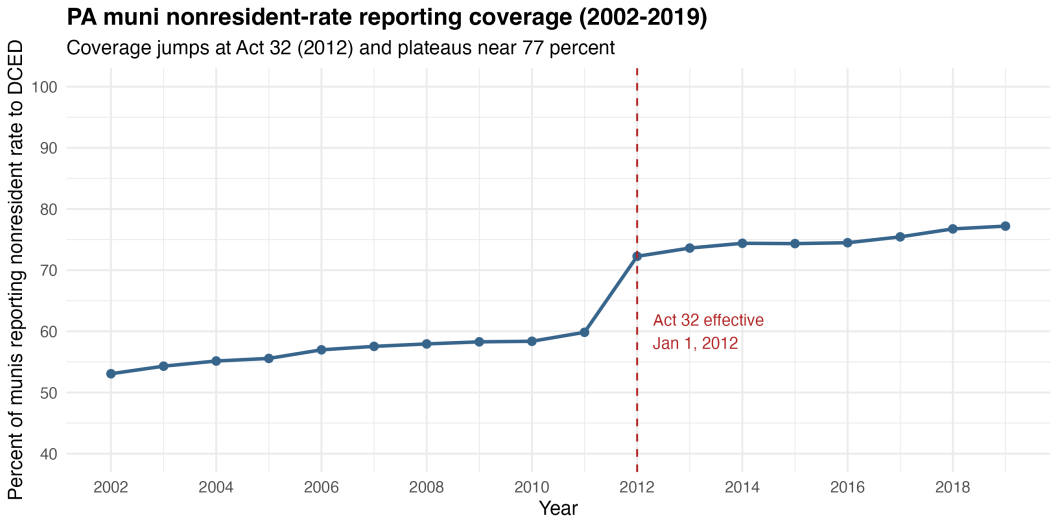



Figure 3: Share of PA municipalities reporting a nonresident rate to DCED, by year. The break at 2012 reflects the Act 32 requirement that employers withhold at the higher of the home resident and work nonresident rates.

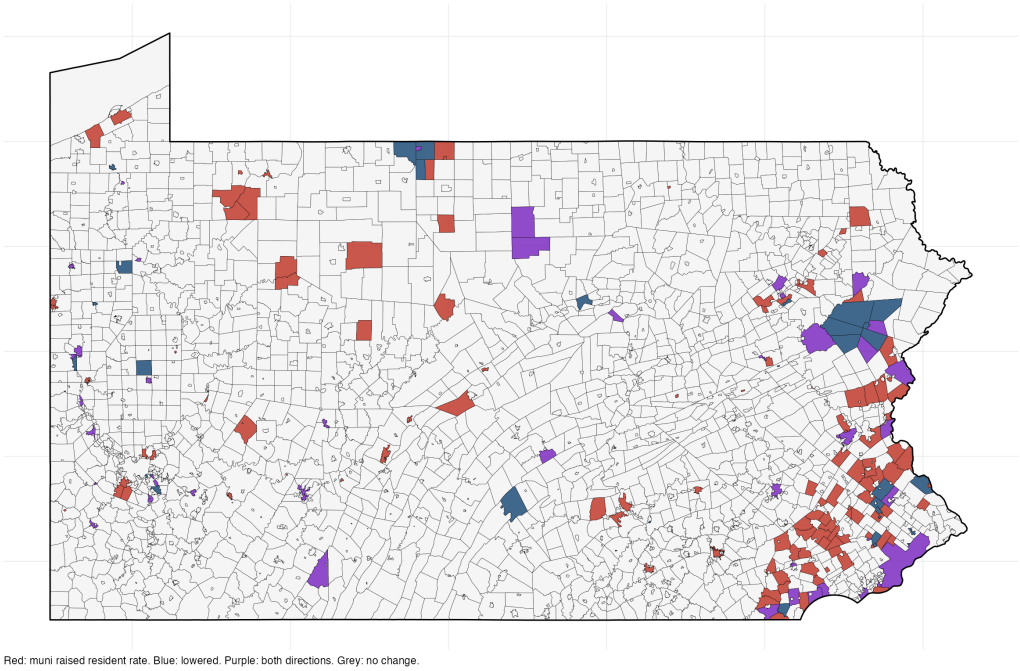



Figure 4: Pennsylvania EIT rate changes, 2000–2019. Red = increased; blue = decreased; purple = both; grey = no change.

### 4.2.1 Cell counts and stacked-design support

Before turning to the empirical strategy, this subsection documents the sample composition of the binding-state decomposition. The purpose is twofold: to verify that each cell in Section **??** has enough municipalities to identify a per-pp coefficient, and to characterize *which* types of Pennsylvania municipalities populate the treated cells—a within-state heterogeneity that matters because PA legal-organization classes (city, borough, township) have different revenue authorities and fiscal-capacity profiles.

Table **??** reports the number of unique municipalities in each binding-state cell on each rate-setter side after the NC-purity filter described in Section **??**.

Table 5: Cell composition by side (drop sample, post-NC-fix)

| Side | Inc-bind | Inc-becomes | Inc-nobind | Dec-bind | Dec-becomes-non | Dec-nobind | NC |
|---|---|---|---|---|---|---|---|
| Muni resident ($\tau^{\text{muni}}$) | 40 | 58 | 26 | 11 | 4 | 15 | 2,406 |
| School district ($\tau^{\text{SD}}$) | 75 | 197 | 14 | 4 | 10 | — | 2,253 |
| Muni nonresident ($\tau^{\text{nr}}$) | 0 | 0 | 171 | 0 | 5 | 109 | 2,276 |

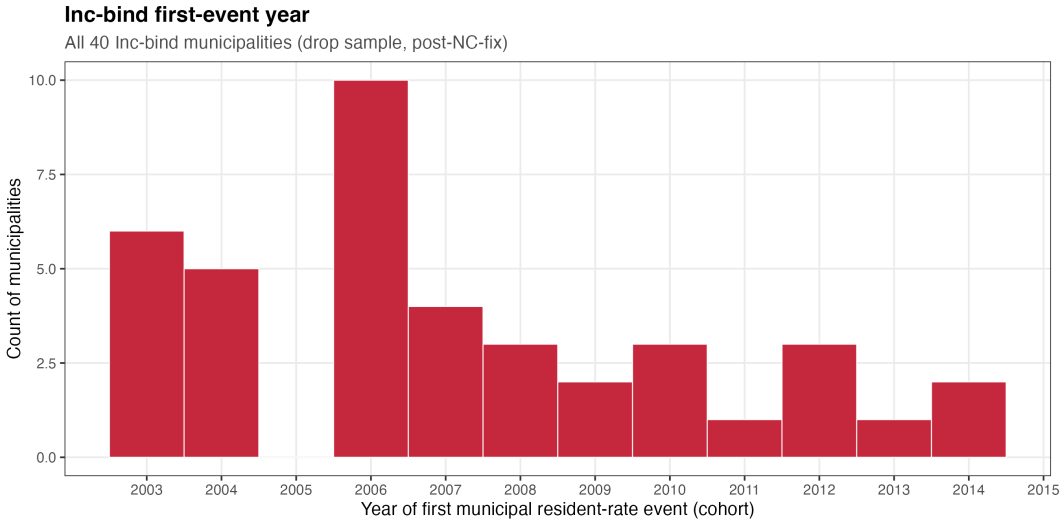
*Notes:* Cells classified at each municipality's first observed rate event on the relevant side. Munis with rate changes whose first-event binding state cannot be determined (NA pre-event nonresident rate) are dropped (not pooled with NC). On the nonresident side, Inc-bind and Inc-becomes cells are empty in PA because the municipality's nonresident rate almost never exceeds the municipality's own home stack at the municipality-self-pair level.

The 40 Inc-bind municipalities, when aggregated, are disproportionately concentrated in older industrial cities and boroughs of Pennsylvania (Wilkes-Barre, Coatesville, Aliquippa, Johnstown, Hazleton). Decrease cells are extremely thin (11 Dec-bind, 4 Dec-becomes-non, 15 Dec-nobind in total), so the binding-rule symmetry test between Inc and Dec is not feasible under conventional standard-error inference.

The stacked design's identifying support comes from the joint distribution of cohort timing and event-time coverage. Figure **??** reports the distribution of municipality-rate first-event years for the 40 Inc-bind municipalities. The cohort distribution is bimodal: an

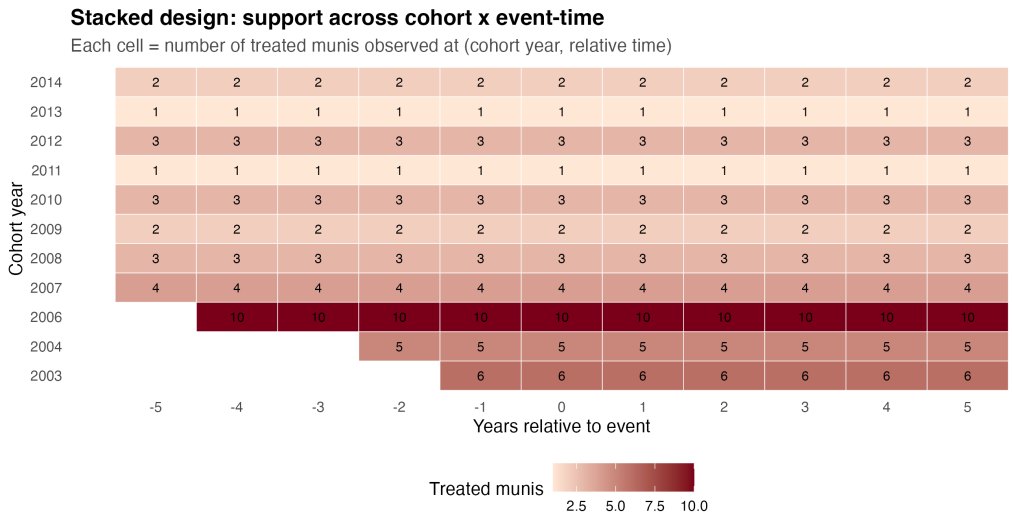
early spike in 2003–2004 (boroughs and townships responding to baseline EIT setup) and a 2006 spike of 10 units (Act 1 of 2006 enabled SD-rate increases that mechanically flipped some municipalities into Inc-bind via the joint $\tau^{\text{muni}} + \tau^{\text{SD}}$ stack), with a long tail through 2014. Figure **??** shows the resulting stacked-design support: the 2006 cohort dominates the post-event window with 10 treated municipalities, and most other cohorts contribute 1–4 treated municipalities to the stack. The thin tail cohorts limit the effective sample size for late-event-time coefficients in each cohort-specific stack.

Figure 5: Inc-bind first-event year.




*Notes:* Counts of Inc-bind municipalities by year of first municipality-rate event, aggregated across municipality classifications. The 2006 spike reflects Act 1 of 2006 (school-district EIT authorization), which mechanically flipped some municipalities into the binding cell via the joint home-stack $\tau^{\text{muni}} + \tau^{\text{SD}}$.

Figure 6: Stacked design support: treated municipality count by cohort year × relative event-time.




*Notes:* Each cell reports the number of Inc-bind treated municipalities observed at the (cohort year, relative event-time $k$) combination. For example, the cell at row $c = 2006$ and column $k = +3$ shows the count of Inc-bind municipalities whose first-event year was 2006, observed in calendar year 2009 (three years after their rate change took effect). Pre-event cells with calendar year before 2002 are empty because LODES coverage begins in 2002, so cohorts at $c = 2003$ and $c = 2004$ have truncated pre-event windows.

# 5 Empirical Strategy

The empirical framework is a continuous stacked difference-in-differences design with binding-state cell decomposition, estimated on a panel spanning 2002–2019 at three units of observation: municipality, school district, and census block. Pre-2002 data are not consistently available in LODES; post-2019 years are excluded because COVID disrupted commuting patterns and ARPA fiscal transfers distorted municipal budgets. Each estimating unit plays a different role. The block-level estimate is the preferred causal specification because both rate-setting jurisdictions enter the regression at their native geography and the realized-dose pre-trend on the binding cell is clean. The municipality-level estimate is retained as policy-jurisdiction magnitude calibration, and the school-district aggregation is used as a

diagnostic. The first-difference dose-response specification estimates the post-event effect of a one percentage-point change in the relevant tax rate; a separate realized-dose event-study diagnostic tests for differential pre-event trajectories across units of varying eventual treatment intensity.

The resident-side specification includes two simultaneous rate-setter treatments: $\Delta\tau_{it}^{\text{muni}}$, the cumulative change in the municipality resident rate from baseline, and $\Delta\tau_{it}^{\text{SD}}$, the cumulative change in the school-district rate. Baseline for each is the rate at year $c_i^s - 1$, where $c_i^s$ is unit $i$'s first event year on side $s \in \{\text{municipality, SD}\}$, and cumulative dose accumulates with subsequent rate changes within $i$. The work-side nonresident rate $\tau^{\text{nr}}$ enters the binding-state classification of cells but is not itself a treatment in the headline regression: I focus on the resident-side mechanism and defer the work-side behavioral response to a separate analysis.

Each rate-change event is classified by direction $\in \{\text{Inc, Dec}\}$ and binding-state transition $\in \{\text{bind, becomes, nobind}\}$ to yield the six non-NC cells introduced in Section **??**. For each unit $i$ and each rate-setter side $s$, I locate the first rate event year $c_i^s$ and assign $i$ to one of the six cells based on the direction of the change and the binding state of the home stack relative to $\tau^{\text{nr,muni}}$ in the pre- and post-event years. The NC reference category is restricted strictly to units whose rate $\tau^s$ never changes over 2002–2019. Units with rate changes whose first event has an undefined binding state—typically because the municipality's nonresident rate is missing in the pre-event year due to DCED reporting gaps before Act 32—are categorized as "Uncertain" and dropped: pooling them with NC would contaminate the control trend with treated-but-unclassified units, while reporting them in their own cell would mix true binding states with measurement gaps. Municipality-level regressions further impose a minimum employment threshold of 50 total jobs. The threshold protects the share-based outcomes (*share_live_in*, *share_work_outside*) from mechanical measurement noise: at $N$ workers, a single worker moving changes the share by $1/N$, so at $N = 10$ a one-worker move changes the outcome by 10 percentage points, dominating

the per-pp treatment magnitude. At $N = 50$ the one-worker noise floor is 2%, roughly an order of magnitude below the headline coefficient. The threshold also matches the LODES synthetic-noise-infusion design: the Census Bureau's noise calibration is validated at the tract level and above (**grahamDESIGNCOMPARISONLODES**), which in Pennsylvania typically contains at least 50 workers. Choosing exactly 50 rather than 20 or 100 is a judgment call; sensitivity analysis in Section **??** confirms the headline Inc-bind coefficient is stable across thresholds from 10 to 500. Imposing the threshold drops 185 of 2,559 PA municipalities (7.2%), almost all of which are sparsely populated rural townships in the NC pool; the treated cohort is essentially unaffected.

Outcomes are run in parallel across counts and shares. The municipality-level workplace and residence outcomes are the logs of `live_and_work` (residents employed in the municipality), `in_commuters` (workers in the municipality from elsewhere), `out_commuters`, `total_jobs`, and `total_residents`, plus the shares of workers living in their own municipality, of workers who are nonresidents, and of residents working outside the municipality. The school-district aggregation uses the same outcomes computed by summing block-level LODES counts up to the SD: each block has a unique cousub assignment (via the TIGER 2020 block-to-cousub crosswalk), each cousub has a primary SD assignment (the modal `sd_id` in the DCED rate panel; 97.6% of munis are single-SD), and the SD-level outcomes are block sums via this two-step block $\rightarrow$ cousub $\rightarrow$ SD path. At the block level, outcomes are the log count of workers living in the block, the log count of workers working in the block, the share of block residents working in their own municipality, and the share working outside; geometry and aggregation details are in Section **??**.

Fiscal controls are share-based and time-varying, with the exact set keyed to the estimating unit. The shares are constructed as numerator-over-total ratios so that they absorb compositional changes in the municipality's or district's fiscal posture without conflating with the level of total spending.

Municipality-level regressions use six share controls from the DCED Annual Financial

Report. Three are expenditure shares: `share_police` (police-services line / total expenditure), `share_streets` (streets and highways), and `share_debt_svc` (debt service: principal and interest payments on outstanding general-obligation and revenue debt). Three are revenue shares: `share_eit` (Earned Income Tax revenue collected / total revenue), `share_re` (real-estate property tax revenue / total revenue), and `share_intergov` (intergovernmental revenue, which is grant and pass-through revenue from state and federal governments—e.g. Liquid Fuels Tax distributions, Community Development Block Grants, ARPA transfers—divided by total revenue).

School-district regressions use eight share controls from the Census Annual Survey of School System Finances (F-33) (**censusF33**). Four are revenue shares: `share_state_rev_sd` (state aid: basic education subsidy, special-education subsidy, transportation reimbursement, etc.), `share_local_rev_sd` (locally raised revenue: property tax, EIT, occupation tax, public-utility taxes), `share_federal_rev_sd` (federal Title I, IDEA, school-lunch, and other direct or pass-through federal aid), and `share_property_tax_sd` (real-estate property-tax revenue as a share of total revenue, which is a subset of `share_local_rev_sd` and isolates the SD's primary local-tax lever). Four are expenditure shares: `share_instructional_sd` (teacher salaries, classroom supplies, instructional staff), `share_support_sd` (non-instructional operating costs: administration, transportation, maintenance, food service, student services), `share_capital_sd` (capital outlay—construction, renovation, equipment), and `share_debt_int_sd` (interest payments on the district's outstanding debt; principal appears as a capital outlay rather than as a current expense).

Block-level regressions use both sets of controls simultaneously, since each block is exposed to both rate-setters' fiscal posture. Figure **??** plots the empirical distribution of each share variable pooled across municipality-years (municipality shares) or SD-years (SD shares); cross-unit variation is broad on every dimension, so the share controls absorb meaningful fiscal heterogeneity without collinearity with the rate-treatment variable.

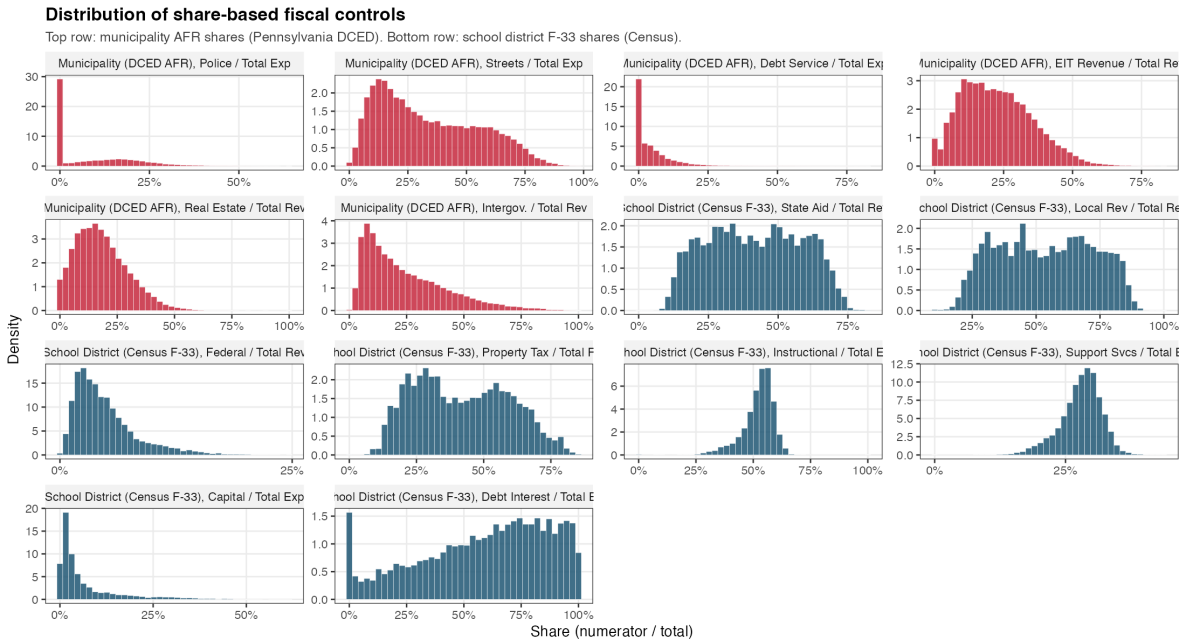



Figure 7: Distribution of share-based fiscal controls across the Pennsylvania panel, 2002–2019. Top six panels: municipality AFR shares (DCED). Lower eight panels: school-district F-33 shares (Census). Each panel pools all municipality-years (top) or SD-years (bottom) over 2002–2019.

## 5.1 Pooled Aggregate Specification

Before the binding-state decomposition I report an aggregate specification that pools all six binding-state cells per side into two direction tags (Inc for any rate increase, Dec for any rate decrease). The pooled regression is the literature-comparable baseline: it answers "what is the average per-pp effect of a municipality-rate hike on residents, ignoring binding state?" This is closest in form to the population-elasticity work of **yangRESPONSIVENESSINCOMELOCAL** on Indiana local income taxes. The binding-state decomposition that follows in Equation (**??**) then explains which sub-population of the treated municipality cohort drives the pooled coefficient.

For each unit-time observation $(i, t)$ on the stacked panel, define the per-pp pooled spec

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} &+ \beta^{M}_{\text{Inc}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Inc}\} \;+\; \beta^{M}_{\text{Dec}}\, \Delta\tau^{M}_{it}\, \mathbf{1}\{d^{M}_{i} = \text{Dec}\} \\ &+ \beta^{SD}_{\text{Inc}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Inc}\} \;+\; \beta^{SD}_{\text{Dec}}\, \Delta\tau^{SD}_{it}\, \mathbf{1}\{d^{SD}_{i} = \text{Dec}\} \\ &+ \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{3}$$

where $d^{s}_{i} \in \{\text{NC}, \text{Inc}, \text{Dec}\}$ is the direction tag on side $s \in \{M, SD\}$ (NC is the omitted reference), $\Delta\tau^{s}_{it}$ is the cumulative dose defined in Section **??**, and the fixed effects and clustering follow the rule in Section **??**: unit-by-stack and year-by-stack fixed effects, with clustering at the policy-jurisdiction level.

The pooled coefficient $\beta^{M}_{\text{Inc}}$ is a weighted average of the six binding-state cell coefficients on the municipality side (Inc-bind, Inc-becomes, Inc-nobind; with equivalent decomposition for Dec). It is interpretable as the elasticity a prior-literature specification would recover. The decomposition in Section **??** then reveals that the pooled coefficient is dominated by the Inc-bind cell at the block level and attenuated toward zero in the Inc-nobind placebo cell—direct evidence that the institutional binding rule, not the statutory rate *per se*, drives the response.

I also report a *binary* analog of Equation (**??**) in which the four per-pp interactions are replaced by four post-treatment indicators $\{\text{everInc}^{M}_{it}, \text{everDec}^{M}_{it}, \text{everInc}^{SD}_{it}, \text{everDec}^{SD}_{it}\}$, each equal to 1 in the event year and after for the relevant direction on the relevant side. The binary coefficients are comparable to the event-study coefficients in Bagchi (**bagchiEffectsIntroducingWitl** both versions appear together in Section **??**.

## 5.2 Binding-State Decomposition: Stacked Continuous DiD

The binding-state decomposition extends the pooled spec by splitting each direction tag into its three binding-state cells, and is the main estimator of the paper. The construction is stacked continuous difference-in-differences. For each rate-setter side $s \in \{\text{muni, SD}\}$ and

each cohort year $c$ at which any municipality's first event on side $s$ occurs, I build a sub-experiment $\mathcal{S}_c^s$ consisting of two types of units. The treated set is the municipalities with first-event year $c_i^s = c$ and a non-NC cell on side $s$; the control set is all NC$^s$ municipalities (whose rate $\tau^s$ never changes over 2002–2019). The window covers calendar years $t \in [c-K,\, c+K]$ with $K = 5$. The choice $K = 5$ is conventional in the staggered-DiD literature and is calibrated against the panel: $K = 5$ yields a 5-pre, 5-post window that fits inside 2002–2019 with full coverage for cohorts $c \in [2007, 2014]$ (i.e. the bulk of the treated mass) and gives the realized-dose ES four pre-event leads to test pre-trends. A shorter $K$ would lose post-event coefficients before the cumulative dose has fully realized; a longer $K$ would truncate the pre-period on early cohorts and the post-period on late cohorts. Sensitivity to $K \in \{3, 4, 6, 7\}$ leaves the headline coefficient stable to two decimal places (not reported separately).

A concrete example helps fix ideas. Wilkes-Barre is an Act 47 distressed city with a municipality resident-rate increase from 1.5% to 2.0% in 2009. Its work-side nonresident rate is 1.0%, so the home stack of $(1.5 + 0.5) = 2.0\%$ pre-event already exceeds the nonresident rate; the home stack continues to bind post-event at $(2.0 + 0.5) = 2.5\% > 1.0\%$. Wilkes-Barre is therefore an Inc-bind municipality with $c_{\text{WB}}^M = 2009$. The sub-experiment $\mathcal{S}_{2009}^M$ runs on the window $[2004, 2014]$. Its treated set includes Wilkes-Barre and any other Inc-bind / Inc-becomes / Inc-nobind / Dec-cell municipality whose first municipality event was 2009 ($\approx$ 10 munis); its control set is the $\sim$2,400 NC munis whose resident rate stayed flat across 2002–2019. Each NC municipality contributes 11 municipality-year observations ($[2004, 2014]$) to this sub-experiment. The dose variable $\Delta\tau_{it}^{\text{muni}} = \tau_{it}^{\text{muni}} - \tau_{i,2008}^{\text{muni}}$ is zero for $t \leq 2008$ and equals 0.005 for $t \geq 2009$ for Wilkes-Barre (and analogously for other 2009 cohort munis based on their own pre-event baseline). Relative event time is $\ell_{it} = t - 2009$, so $\ell = -5, -4, \ldots, 0, 1, \ldots, 5$.

Each municipality-year observation in $\mathcal{S}_c^s$ carries a stack indicator $s_i = c$, relative event time $\ell_{it} = t - c$, and stack-by-municipality and stack-by-year identifiers $\tilde{i} = (s_i, i)$ and $\tilde{t} = (s_i, t)$ that allow each unit to appear in the stacked panel multiple times under different

stack identifiers (this is how NC controls re-appear across cohorts without contaminating cohort-specific identification). Stacking across all cohort years yields a single panel in which each NC municipality appears once per cohort and each treated municipality appears in exactly one sub-experiment—its first-event cohort. Within each sub-experiment, controls are clean (no treated municipality from another cohort serves as control), which is the property that fixes the negative-weighting problem in vanilla TWFE under staggered treatment (**goodman-baconDifferenceindifferencesVariationTreatment2021**).

Figure **??** sketches the construction. Each row is one example municipality. The colored bar is its $[c-K, c+K]$ window; the open circle marks its first-event year $c$; the cell label on the right is the binding-state classification. The six treated examples are one per binding-state cell. The bottom row is the never-changed (NC) control band: every NC municipality contributes observations to *every* treated cohort's sub-experiment, in calendar years matching that cohort's window. Early cohorts (e.g. the 2006 Inc-becomes example) have a slightly truncated pre-period because their $c-K$ year falls just before the panel's 2002 start; the stacked panel handles this by simply omitting the unobserved pre-event years.

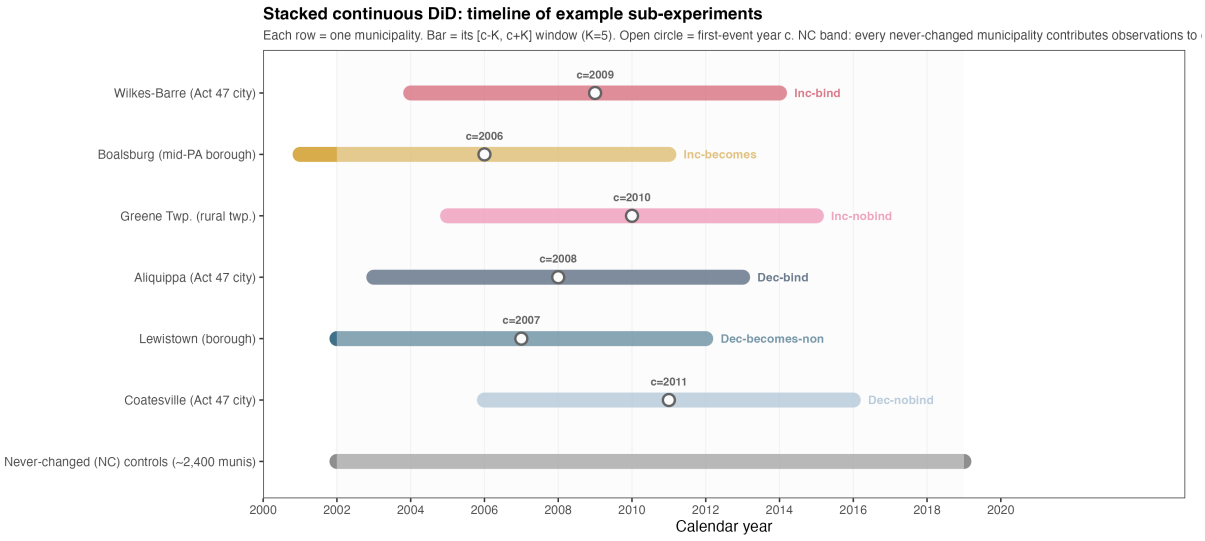



Figure 8: Stacked continuous DiD: timeline of example sub-experiments. Each row is a municipality; the colored bar is the $[c-K,\ c+K]$ window with $K = 5$; the open circle marks the first-event year $c$; the cell label gives the binding-state classification. The bottom (grey) row is the never-changed (NC) control pool; in the stacked panel, every NC municipality contributes observations to every cohort's window. Municipality names are illustrative.

The stacked-DiD construction follows the negative-weighting fix of **goodman-baconDifferenceindiffe** the binding-state cell partitioning is novel and central to the design's within-design placebo (Inc-nobind), which the standard stacked DiD cannot produce.

Some treated municipalities change rates more than once in 2002–2019, raising a coding question about which event defines the cohort. A treated municipality $i$ may have more than one rate-change event over the panel. I classify each such municipality by the binding-state transition of its first event and assign $i$ to the corresponding cell on side $s$. The municipality then appears in exactly one sub-experiment (its first-event cohort $c_i^s$), not in subsequent cohorts triggered by later events.

Subsequent rate changes within the same municipality accumulate into $\Delta\tau_{it}^s$, the *cumulative* dose from the municipality's pre-event baseline. Concretely: suppose municipality $i$ has $\tau_i^s = 0.50$ in years 2005–2009, $\tau_i^s = 0.70$ in 2010 (first event, $c_i^s = 2010$, baseline $= 0.50$), and $\tau_i^s = 1.00$ from 2014 onward (second event). Then $\Delta\tau_{it}^s = 0$ for $t < 2010$, $\Delta\tau_{it}^s = 0.20$ for $t \in [2010, 2013]$, and $\Delta\tau_{it}^s = 0.50$ for $t \geq 2014$. The stacked-continuous specification absorbs

this trajectory through the dose-stack interaction in a single regression on municipality $i$'s 2005–2015 window (its first-event cohort 2010 with $K = 5$). Muni $i$ is *not* duplicated as a 2014 cohort treated municipality; it stays in its 2010 cohort. The cumulative dose path encodes the second event by raising $\Delta\tau^s_{it}$ from 0.20 to 0.50 at year 2014.

This first-event classification has three deliberate consequences. First, no treated municipality serves as control to itself; each municipality appears in exactly one stack. Second, multi-event treated municipalities do not receive disproportionate weight from appearing in multiple stacks. Third, the per-pp coefficient $\beta^s_k$ is interpretable as a dose-response elasticity over the cumulative rate change, which is the welfare-relevant magnitude for fiscal calibration.

For each treated municipality $i$ in stack $\mathcal{S}^s_c$, define the cumulative dose-stack as $\Delta\tau^s_{it} = \tau^s_{it} - \tau^s_{i,c-1}$, with $\Delta\tau^s_{it} = 0$ for $t < c$ (pre-event) and for NC municipalities at all $t$. The stacked continuous DiD specification, for outcome $Y$, runs the regression on the stacked panel:

$$\begin{aligned} Y_{\tilde{i}\tilde{t}} \;=\; \alpha_{\tilde{i}} \;+\; \gamma_{\tilde{t}} + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{muni}}_k \, \Delta\tau^{\mathrm{muni}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{muni}}_i = k\} \\ + \sum_{k \in \mathcal{C}_{\setminus \mathrm{NC}}} \beta^{\mathrm{SD}}_k \, \Delta\tau^{\mathrm{SD}}_{it} \cdot \mathbf{1}\{\mathrm{cell}^{\mathrm{SD}}_i = k\} \\ + \mathbf{X}'_{it}\boldsymbol{\phi} \;+\; \varepsilon_{\tilde{i}\tilde{t}}, \end{aligned} \tag{4}$$

where $\mathcal{C}_{\setminus \mathrm{NC}} = \{$Inc-bind, Inc-becomes, Inc-nobind, Dec-bind, Dec-becomes-non, Dec-nobind$\}$ is the set of non-NC cells. NC is the omitted reference category on each side, so for each of the two rate-setter sides (municipality and SD) the regression estimates six $\beta_k$ coefficients, each interpreted as the dose-response coefficient relative to the NC trend on that side, holding the other side's cell $\times$ dose interaction fixed. The unit of observation $\tilde{i}$ in the regression is a (municipality, stack) pair: each treated municipality contributes observations in exactly one stack (its first-event cohort), while each NC municipality contributes observations in every stack. The fixed effect $\alpha_{\tilde{i}}$ is therefore a separate intercept for each (municipality, stack)

combination—so a single NC municipality has a different FE in each cohort it appears in—and $\gamma_{\tilde{t}} = \gamma_{(c,t)}$ is a separate calendar-year shock per cohort. The control vector $\mathbf{X}_{it}$ holds the share-based fiscal controls described above. Standard errors are clustered at the underlying *municipality*, not at the (municipality, stack) level. Cluster-at-municipality is the right choice because the same municipality's outcomes are correlated across the multiple stacks in which it appears as an NC control; the municipality-by-stack FE absorbs the level but not the residual covariance.

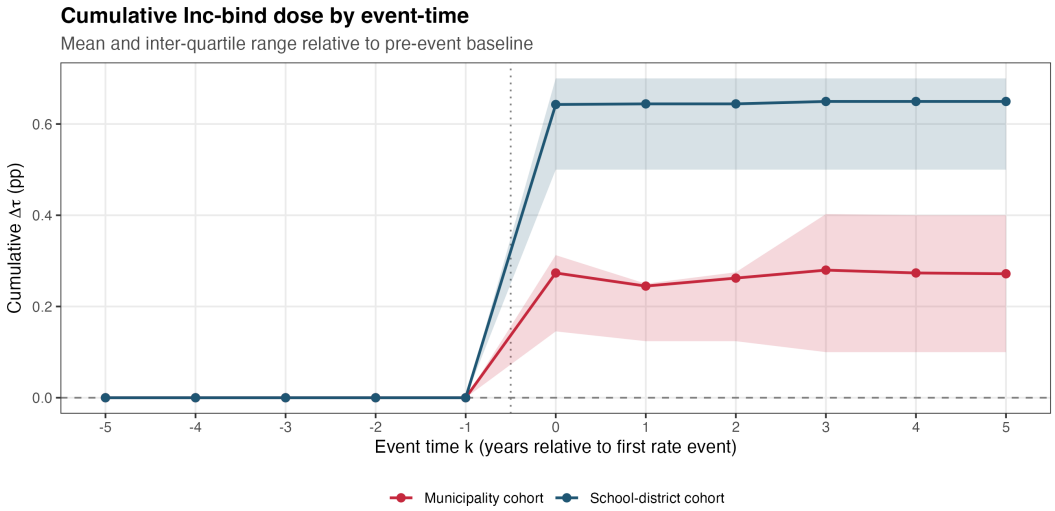
The cell coefficient $\beta_k^s$ is the per-pp dose-response of outcome $Y$ to a 1 percentage-point change in rate $s$, restricted to municipalities whose first event on side $s$ falls in cell $k$, with NC municipalities on side $s$ as controls. Specifically, $\beta^{\text{muni}}_{\text{Inc-bind}}$ is the dose-response of $Y$ to municipality-rate hikes among municipalities whose home stack was binding both pre- and post-event, identified by comparing their post-event trajectory to NC municipalities' contemporaneous trajectory. The placebo within the design is $\beta^{\text{muni}}_{\text{Inc-nobind}}$: under the institutional rule, this should be zero (the rate hike does not move the bill).

Within each sub-experiment, the comparison is treated cohort vs NC controls in the same calendar-year window. The cell decomposition partitions the treated set by whether the rate change moved the bill. The reference for each cell coefficient is the NC trend on the same side. The triple decomposition (municipality $\times$ SD $\times$ nr) lets us read off, in a single regression: which rate moves which outcome, conditional on the municipality's binding state on that side.

The identifying variation in this design comes from the cross-unit distribution of cumulative rate changes after each unit's first event. Figure **??** traces the mean cumulative municipality-resident-rate change (with IQR) of Inc-bind units relative to their pre-event baseline. The pre-event window is mechanically flat at zero by construction (dose is measured relative to the unit's own $c-1$ rate); the post-event window shows the realized cumulative $\Delta\tau^{\text{muni}}$ rising to roughly 0.2 percentage points for townships, 0.3 for boroughs, and 0.4 for cities by $k = +5$. Figure **??** reports the cross-unit distribution of the cumulative dose at

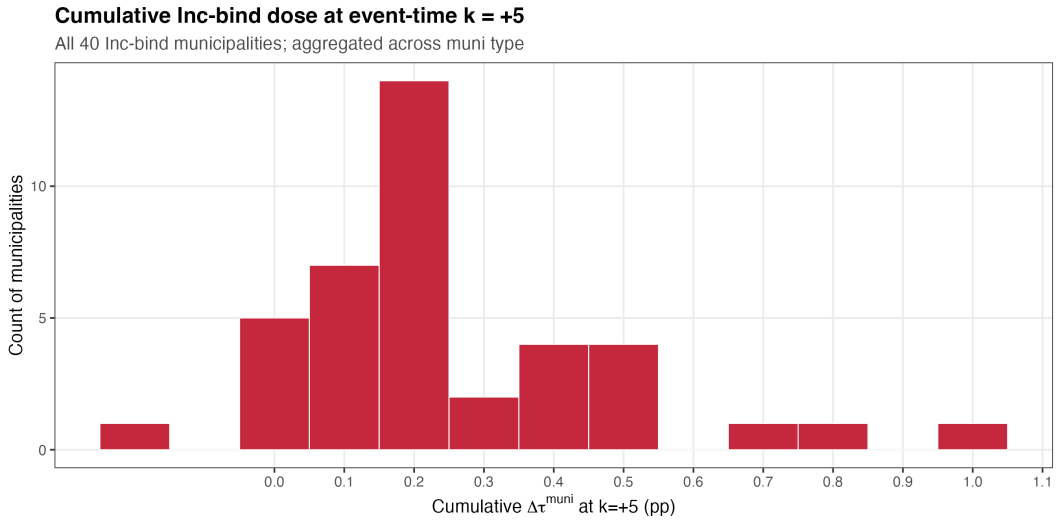
$k = +5$: most Inc-bind municipalities experience cumulative changes of 0.2–0.4 pp, with a small right tail (Wilkes-Barre and Coatesville) at 0.8–0.9 pp. The realized-dose ES diagnostic exploits this across-unit variation to test pre-trends conditional on eventual treatment intensity.

Figure 9: Cumulative Inc-bind dose by event-time: municipality cohort vs. school-district cohort.




*Notes:* Mean and inter-quartile range of cumulative dose by relative event-time $k$, separately for the municipality cohort (red, $\Delta\tau_{it}^{\text{muni}}$ relative to baseline $\tau_{i,c-1}^{\text{muni}}$) and the school-district cohort (blue, $\Delta\tau_{it}^{\text{SD}}$ relative to $\tau_{i,c-1}^{\text{SD}}$). Pre-event values are zero by construction.

Figure 10: Distribution of cumulative Inc-bind dose at $k = +5$.




*Notes:* Cross-municipality distribution of cumulative resident-rate change at the end of the post-event window ($k = +5$), aggregated across municipality classifications. The mass at 0.2–0.4 pp reflects the standard half-point increment; the right tail at 0.8–0.9 pp is driven by Home Rule cities.

## 5.3 Three Estimating Units

The headline equation (**??**) is estimated at three units of observation. The choice of unit interacts directly with how the two rate-setting bodies (municipality and school district) enter the regression.

At the municipality level, the unit of observation is municipality × year. Workplace-panel and residence-panel outcomes are constructed by aggregating LODES origin–destination flows over blocks within the municipality. The municipal resident rate enters the regression at its native level. The school district resident rate is assigned to each municipality-year as $\max(\tau^{SD})$ across the school districts the municipality reports to in DCED records; for the 97.6% of municipalities that report to a single SD this is exact, and for the 2.4% that span multiple SDs the max-aggregation is conservative. Fiscal share controls are taken from the municipality AFR. Fixed effects: municipality-by-stack and year-by-stack. Standard errors clustered at the municipality level.

At the school district level, the unit is SD × year. Outcomes are constructed by aggregating LODES OD flows over blocks within the SD. The SD resident rate enters at its native level. The municipal resident rate is assigned to each SD-year as the within-SD aggregation across the SD's constituent municipalities. Fiscal share controls are taken from Census F-33 SD-level revenue and expenditure data, mirroring the municipality AFR shares. Fixed effects: SD-by-stack and year-by-stack. Standard errors clustered at the SD level.

At the block level, the unit is census block × year. Each block is assigned its own municipality and its own SD, so both rate-setters' policy variation enters the regression at its native level with no within-jurisdiction aggregation. Outcomes are block-defined: log of workers living in the block, log of workers working in the block, the share of block residents who work in the block's own municipality, and the share of block residents who work outside the municipality. Fiscal share controls include both municipality AFR shares and F-33 SD shares, since each block is exposed to both governments. Fixed effects: block-by-stack and year-by-stack. Standard errors two-way clustered at the municipality and SD levels.

The block level is the preferred causal estimate for two reasons. First, the block is the only level where both rate-setting jurisdictions enter at their native geography, with no within-jurisdiction aggregation step the reader has to defend. Second, the realized-dose pre-trend diagnostic is cleanest at the block level (Table **??**): all four pre-event coefficients are individually $p > 0.10$, with a mean pre-period coefficient of $-0.027$. The block-level joint Wald F-test rejects parallel trends ($p < 0.001$) but the rejection is driven by the test's enormous statistical power at $N \approx 21$M block-years rather than by economically meaningful deviations. The municipality-level analogue on the log count of residents working in their own municipality returns a joint Wald F-test that does *not* reject parallel trends ($p = 0.23$, stat $= 5.62$ on 4 df), with three of four pre-period coefficients individually $p > 0.10$. The municipality-level diagnostic is therefore the cleanly-non-rejecting one, while the block-level diagnostic is hyper-powered rather than a sharp failure, but it is enough that I treat the block-level estimate as the causal headline and the municipality-level as

policy-jurisdiction magnitude calibration: the typical PA municipality has ∼40 live-and-work workers, a meaningful per-jurisdiction benchmark, but the noisier pre-period in the diagnostic means I do not promote the municipality-level Inc-bind coefficient to a second causal headline. The school-district aggregation produces direction-mixed coefficients that do not survive the pre-trend diagnostic and is reported as an aggregation diagnostic in the Companion Appendix.

## 5.4 Pre-Trend Diagnostics

Because the first-difference dose $\Delta\tau_{it}$ is mechanically zero in all pre-event periods by construction, the headline specification cannot estimate conventional pre-treatment lead coefficients. I report a realized-dose event-study diagnostic that tests *strong* parallel trends across treatment-intensity magnitudes, as recommended by the recent continuous-DiD literature for designs with a continuous dose.

### 5.4.1 Realized-Dose Sun–Abraham Event Study

To assess parallel trends in the cross-dose sense (**callaway2021**; **dechaisemartin2024**), I estimate a separate event-study specification using each unit's realized cumulative dose, $\widetilde{D}_i = \tau^M_{i,c+5} - \tau^M_{i,c-1}$, which is constant within unit and varies across units in proportion to eventual treatment intensity. For each municipality-side cell $c$, restrict to the subset of units in $\{\text{NC}, c\}$ and run

$$Y_{it} = \alpha_{\tilde{i}} + \gamma_{\tilde{t}} + \sum_{k \neq -1} \beta^M_k \, \mathbf{1}\{\ell_{it} = k\} \, \widetilde{D}_i + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}, \tag{5}$$

where $\ell_{it} = t - c^M_i$ is event-time relative to the unit's first rate event. The $\beta^M_k$ at pre-event leads ($k \leq -2$) test whether units with larger eventual rate changes were on differentially sloped pre-event trajectories; under parallel trends in the strong sense, these coefficients are zero. The post-event coefficients ($\beta^M_k$ for $k \geq 0$) recover the per-pp dose-response at each

event time and average to the corresponding cell coefficient in the headline specification when the unit's rate path is a single step at the event year.

I do not impose a uniform pre-trend standard across every outcome in the paper, because not every outcome carries the causal claim. The primary residential-sorting outcome is the block-level log count of workers living in the block, and the realized-dose pre-trend on the binding cell must be clean here for the headline to go through; this is the outcome the paper asks the reader to believe. Municipality-level resident counts—log residents working in their own municipality, log employed residents—are retained as policy-jurisdiction magnitude calibration. I acknowledge and caveat imperfect pre-trends at the municipality scale, but these outcomes do not carry the causal claim. Secondary equilibrium outcomes such as log in-commuters, log total jobs, and the industry decomposition are reported as suggestive evidence of a local-services demand channel; imperfect pre-trends are acceptable there because these outcomes are downstream of the residential-sorting margin and are not the identification object. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are sensitive to sparse denominators and to endogenous denominator dynamics at block geography, so I report them in the Appendix and do not lean on them. The headline block-level log home count in the Inc-bind cell passes the strong standard; other outcomes are retained with the caveats appropriate to their role.

## 5.5 Sample Variants

The DCED nonresident-rate panel has reporting gaps—55–60% coverage pre-Act 32, 72–80% post—which affect the binding-state classification when the nonresident rate is missing. I report three sample variants. The strict variant drops municipality-years with no recorded nonresident rate. The filled variant imputes `muni_nonres_eit` as the contemporaneous resident rate when missing, which is statute-anchored: under Act 511 the nonresident rate is bounded above by the resident rate. The drop-incomplete variant keeps all municipality-

years and drops NA only at regression time per outcome, maximizing $N$. The headline tables use the drop-incomplete sample; the other two are reported as robustness columns, and coefficients are stable across them.

Section **??** reports the headline binding-state decomposition at each estimating unit, the pooled Inc/Dec literature benchmark, and the industry-mechanism decomposition. Auxiliary diagnostics (pre/post Act 32 split and sample variants) are in Section **??** and Appendix **??**.

# 6 Results

I report results in two layers. Section **??** presents the pooled Inc/Dec benchmark—what a conventional local-tax design would recover if it ignored the Pennsylvania max-rule. Section **??** reports the binding-state decomposition at three levels of aggregation. The block-level decomposition is the preferred causal estimate because both rate-setting jurisdictions enter at their native geography; the municipality-level decomposition is reported as policy-jurisdiction magnitude calibration; the school-district aggregation is a diagnostic non-result. The central empirical comparison throughout is between the Inc-bind cell (where the statutory rate change moves the worker's bill) and the Inc-nobind cell (where it does not), with Inc-nobind serving as the within-design placebo. Mechanism evidence from the industry decomposition appears in §**??**; the pre-/post-Act-32 enforcement-break split appears in the Robustness section.

## 6.1 Pooled Inc/Dec Benchmark

I first report the pooled Inc/Dec specification from Equation (**??**), which collapses each rate-setter side to two direction tags. The pooled coefficients are useful because they approximate the estimand in a conventional local-tax design: the average per-pp dose-response of a municipality-rate change and an SD-rate change without distinguishing whether the rate change moves the worker's bill. In Pennsylvania, however, the pooled specification is not the

structural object of interest. It averages together rate changes that raise tax liability (Inc-bind) and rate changes that do not (Inc-nobind), and the binding-state decomposition in Section **??** shows which sub-population of the pooled treated cohort drives each coefficient. Cell counts and stacked-design support are reported in Section **??**.

### 6.1.1 Block level

Table **??** reports the per-pp pooled coefficients at the block level. The table includes four coefficients of interest per outcome: the municipality resident-rate Inc effect $\beta^{M}_{\text{Inc}}$, the municipality resident-rate Dec effect $\beta^{M}_{\text{Dec}}$, and the analogous school-district-rate coefficients $\beta^{SD}_{\text{Inc}}, \beta^{SD}_{\text{Dec}}$.

Table 6: Pooled Inc/Dec headline — block level (per pp)

| Outcome | Log(workers live) | Log(workers work) | Share live+work in muni |
|---|---|---|---|
| **Municipality rate ($\tau^M$)** | | | |
| $\Delta\tau > 0$ (Inc) | -0.045 | -0.066*** | -0.041*** |
| | (0.034) | (0.024) | (0.009) |
| $\Delta\tau < 0$ (Dec) | 0.255*** | 0.097 | -0.025 |
| | (0.089) | (0.229) | (0.022) |
| **School district rate ($\tau^{SD}$)** | | | |
| $\Delta\tau > 0$ (Inc) | 0.035 | 0.028 | -0.000 |
| | (0.031) | (0.025) | (0.005) |
| $\Delta\tau < 0$ (Dec) | 0.001 | -0.002 | -0.008*** |
| | (0.008) | (0.011) | (0.003) |
| $N$ | 23,567,655 | 8,751,581 | 23,567,655 |
| $R^2$ | 0.858 | 0.865 | 0.563 |
| Within $R^2$ | 0.000 | 0.000 | 0.000 |
| Dep. mean | 2.238 | 2.318 | 0.121 |
| Dep. SD | 1.186 | 1.655 | 0.184 |

*Notes:* Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are $\beta^M_{\text{Inc}}$, $\beta^M_{\text{Dec}}$, $\beta^{SD}_{\text{Inc}}$, $\beta^{SD}_{\text{Dec}}$. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### 6.1.2 Municipality level

Table **??** reports the analogous per-pp pooled coefficients at the municipality level, with the same four coefficients of interest per outcome.

Table 7: Pooled Inc/Dec headline — municipality level (per pp)

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Municipality rate ($\tau^M$)** | | | | |
| $\Delta\tau > 0$ (Inc) | -0.228*** | -0.209*** | -0.035 | -0.019** |
| | (0.058) | (0.056) | (0.033) | (0.008) |
| $\Delta\tau < 0$ (Dec) | 0.260** | 0.014 | 0.058 | 0.022 |
| | (0.107) | (0.143) | (0.138) | (0.019) |
| **School district rate ($\tau^{SD}$)** | | | | |
| $\Delta\tau > 0$ (Inc) | 0.128* | 0.105 | 0.051 | 0.007 |
| | (0.070) | (0.067) | (0.038) | (0.005) |
| $\Delta\tau < 0$ (Dec) | -0.310*** | -0.170** | 0.020 | -0.003 |
| | (0.110) | (0.076) | (0.051) | (0.007) |
| $N$ | 218,677 | 239,023 | 242,722 | 239,033 |
| $R^2$ | 0.964 | 0.959 | 0.981 | 0.823 |
| Within $R^2$ | 0.001 | 0.001 | 0.001 | 0.001 |
| Dep. mean | 3.677 | 5.847 | 6.697 | 0.093 |
| Dep. SD | 1.934 | 1.887 | 1.299 | 0.076 |

*Notes:* Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are $\beta^M_{\text{Inc}}$, $\beta^M_{\text{Dec}}$, $\beta^{SD}_{\text{Inc}}$, $\beta^{SD}_{\text{Dec}}$. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### 6.1.3 School-district level (brief)

The school-district-level analogue (pooled across the SD's constituent municipality rates) is reported in Table **??**; SD-level identification is weaker than at the municipality or block level and the table is included for completeness.

Table 8: Pooled Inc/Dec headline — school district level (per pp)

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Municipality rate ($\tau^M$)** | | | | |
| $\Delta\tau > 0$ (Inc) | 0.017 | 0.013 | 0.014 | 0.000 |
| | (0.022) | (0.017) | (0.014) | (0.001) |
| $\Delta\tau < 0$ (Dec) | -0.017*** | -0.010* | -0.005*** | -0.000 |
| | (0.003) | (0.005) | (0.002) | (0.000) |
| **School district rate ($\tau^{SD}$)** | | | | |
| $\Delta\tau > 0$ (Inc) | 0.150** | 0.154** | 0.196*** | 0.001 |
| | (0.070) | (0.069) | (0.054) | (0.004) |
| $\Delta\tau < 0$ (Dec) | -0.008 | 0.089 | 0.139*** | -0.012 |
| | (0.069) | (0.078) | (0.040) | (0.010) |
| $N$ | 51,440 | 51,474 | 51,474 | 51,474 |
| $R^2$ | 0.986 | 0.987 | 0.987 | 0.953 |
| Within $R^2$ | 0.003 | 0.002 | 0.004 | 0.004 |
| Dep. mean | 6.260 | 8.335 | 8.768 | 0.125 |
| Dep. SD | 1.298 | 1.202 | 0.909 | 0.068 |

*Notes:* SD-level regression: $\tau^{SD}$ is the school-district's own rate; the muni-rate row uses the within-SD mean of constituent muni resident rates as an aggregated treatment. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are $\beta^M_{\text{Inc}}$, $\beta^M_{\text{Dec}}$, $\beta^{SD}_{\text{Inc}}$, $\beta^{SD}_{\text{Dec}}$. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### 6.1.4 Realized-dose ES benchmark diagnostics

Tables **??** and **??** report the ATT-summary statistics for the block-level pooled event study, separately on the municipality-cohort and SD-cohort stacks, with cluster-robust Wald pre-trend $p$-values. Figures **??** and **??** plot the event-time paths for each stack, side by side, with color encoding direction (red Inc, blue Dec). Tables **??** and **??** report the analogous ATT summaries for the municipality- and SD-level pooled specifications; Figures **??** and **??** report the municipality-level and SD-level pooled event studies separately (red Inc, blue Dec). The Wald $p$-values are the joint pre-trend test $H_0$: $\beta_k = 0$ for all $k \leq -2$ using the model's cluster-robust variance matrix.

Table 9: Pooled realized-dose event study — block level (muni-cohort).

| Direction | Log(workers live) | Log(workers work) | Share live+work in muni |
|---|---|---|---|
| **Inc** (post) | -0.032 | -0.053 | -0.035 |
| | (0.021) | (0.014) | (0.005) |
| **Inc** (pre) | 0.079 | 0.028 | 0.008 |
| | (0.020) | (0.011) | (0.005) |
| **Inc** Wald $p$ | 0.0019 | 0.2118 | 0.0002 |
| **Dec** (post) | 0.217 | 0.123 | -0.045 |
| | (0.046) | (0.100) | (0.007) |
| **Dec** (pre) | 0.171 | 0.160 | 0.018 |
| | (0.052) | (0.118) | (0.010) |
| **Dec** Wald $p$ | $<0.0001$ | $<0.0001$ | 0.0021 |

*Notes:* Cluster-robust standard errors in parentheses. Post = mean of $\beta_k$ for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald $p$ is the cluster-robust joint test of $H_0$: $\beta_k = 0$ across pre-event $k$. Stack built on muni-cohort years; event time defined against each block's muni-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The SD-cohort analogue at the block level uses the same ATT summary statistics with relative time measured against the first SD-rate event.

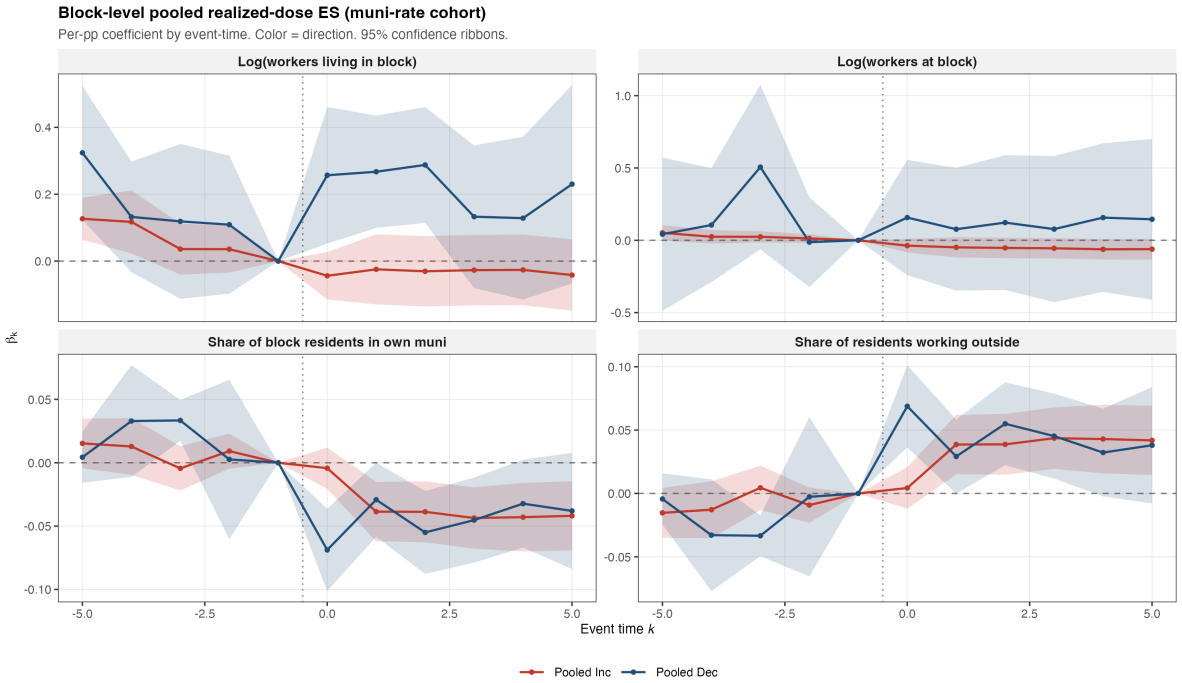
Table 10: Pooled realized-dose event study — block level (SD-cohort).

| Direction | Log(workers live) | Log(workers work) | Share live+work in muni |
|---|---|---|---|
| **Inc** (post) | 0.022 | -0.012 | — |
| | (0.017) | (0.014) | — |
| **Inc** (pre) | -0.126 | -0.003 | — |
| | (0.040) | (0.047) | — |
| **Inc** Wald $p$ | 0.0573 | 0.0151 | — |
| **Dec** (post) | — | 0.322 | -0.117 |
| | — | (0.063) | (0.011) |
| **Dec** (pre) | — | 0.540 | -0.012 |
| | — | (0.131) | (0.008) |
| **Dec** Wald $p$ | — | $<0.0001$ | 0.0067 |

*Notes:* Cluster-robust standard errors in parentheses. Post = mean of $\beta_k$ for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald $p$ is the cluster-robust joint test of $H_0$: $\beta_k = 0$ across pre-event $k$. Stack built on SD-cohort years; event time defined against each block's SD-rate event. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

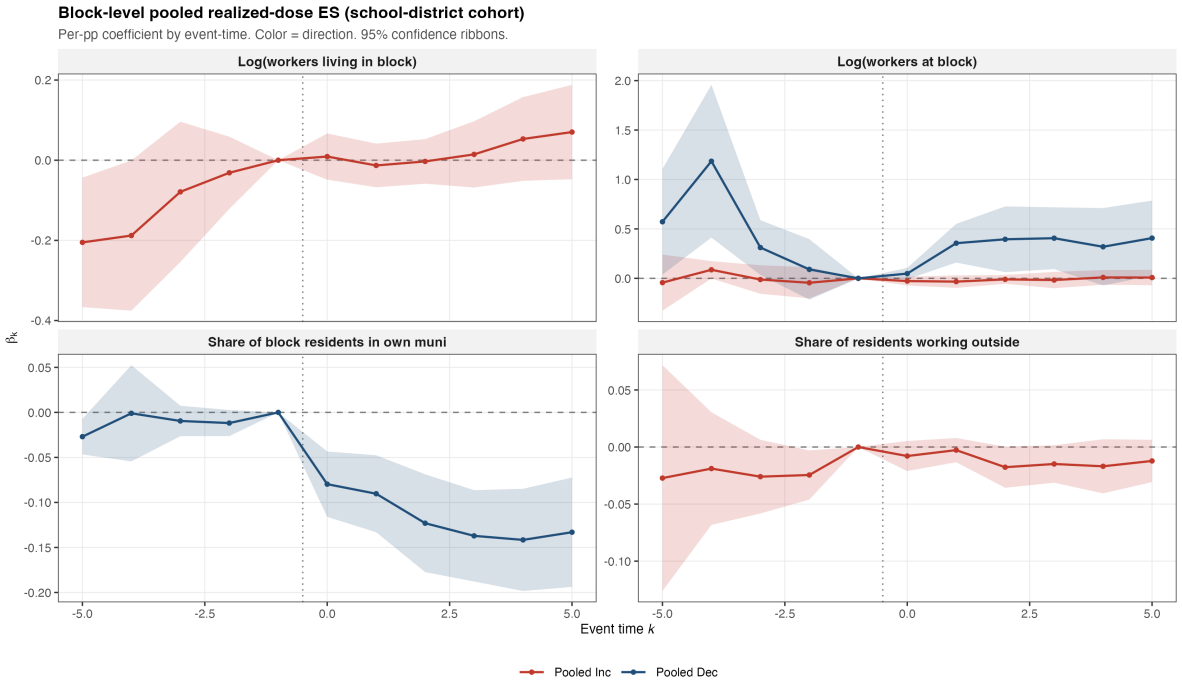
Figure **??** plots the event-time path for the municipality-cohort stack; Figure **??** plots the SD-cohort stack on the same four block outcomes.

Figure 11: Block-level pooled realized-dose event study, municipality-cohort stack. Inc (red) and Dec (blue) coefficients on the four block outcomes corresponding to the four pooled coefficients in Table **??**.




*Notes:* Per-pp realized-dose coefficient at each event-time. Block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects and two-way clustering at the municipality and school-district levels. Relative event-time $k$ is defined against each block's first municipality-rate event year $c_b^M$. Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 12: Block-level pooled realized-dose event study, SD-cohort stack. Inc (red) and Dec (blue) coefficients on the four block outcomes corresponding to the four pooled coefficients in Table **??**.




*Notes:* As in Figure **??**, but with relative event-time $k$ defined against each block's first school-district-rate event year $c_b^{SD}$.

Table 11: Pooled realized-dose event study — municipality level.

| Direction | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc** (post) | -0.197 | -0.218 | -0.039 | -0.015 |
| | (0.026) | (0.028) | (0.014) | (0.003) |
| **Inc** (pre) | 0.094 | 0.010 | 0.054 | 0.011 |
| | (0.034) | (0.058) | (0.032) | (0.006) |
| **Inc** Wald $p$ | $<0.0001$ | 0.0513 | 0.4834 | 0.5071 |
| **Dec** (post) | 0.674 | 0.264 | 0.175 | 0.020 |
| | (0.178) | (0.091) | (0.041) | (0.008) |
| **Dec** (pre) | 0.190 | -0.139 | 0.169 | 0.004 |
| | (0.217) | (0.100) | (0.044) | (0.007) |
| **Dec** Wald $p$ | $<0.0001$ | $<0.0001$ | 0.0003 | 0.0001 |

*Notes:* Cluster-robust standard errors in parentheses. Post = mean of $\beta_k$ for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald $p$ is the cluster-robust joint test of $H_0$: $\beta_k = 0$ across pre-event $k$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The school-district aggregation of the same pooled ES is reported next as a coarser-geography benchmark.

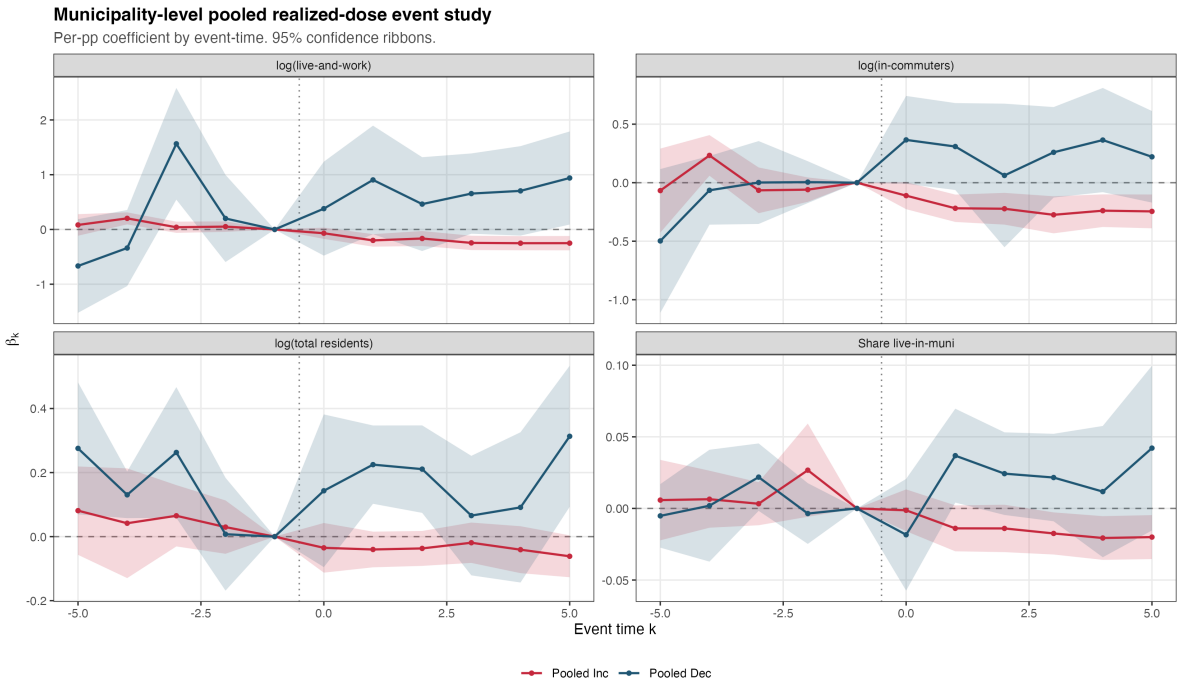
Table 12: Pooled realized-dose event study — school district level.

| Direction | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc** (post) | 0.225 | 0.204 | 0.234 | 0.000 |
| | (0.051) | (0.045) | (0.038) | (0.002) |
| **Inc** (pre) | 0.134 | 0.094 | 0.061 | -0.003 |
| | (0.060) | (0.052) | (0.042) | (0.004) |
| **Inc** Wald $p$ | 0.4556 | 0.2738 | 0.8816 | 0.8688 |
| **Dec** (post) | -0.088 | 0.071 | 0.126 | -0.016 |
| | (0.030) | (0.038) | (0.028) | (0.009) |
| **Dec** (pre) | -0.214 | -0.334 | -0.003 | 0.021 |
| | (0.030) | (0.038) | (0.029) | (0.009) |
| **Dec** Wald $p$ | $<0.0001$ | $<0.0001$ | $<0.0001$ | $<0.0001$ |

*Notes:* Cluster-robust standard errors in parentheses. Post = mean of $\beta_k$ for $k = 0, \ldots, 5$; pre = mean for $k \leq -2$. Wald $p$ is the cluster-robust joint test of $H_0$: $\beta_k = 0$ across pre-event $k$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

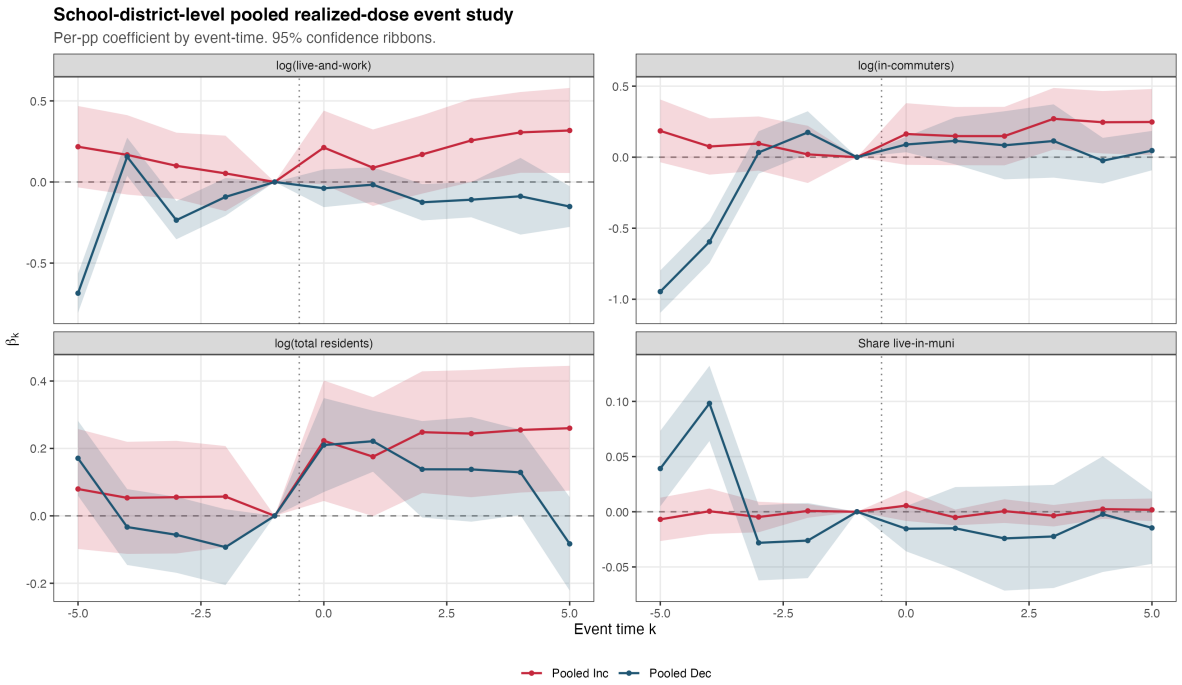
Figures **??** and **??** plot the municipality-level and SD-level pooled event studies separately; the municipality-level Inc post-event path runs monotonically negative from $k = +1$ while the SD-level path is direction-mixed.

Figure 13: Municipality-level pooled realized-dose event study.




*Notes:* Per-pp realized-dose coefficient at each event-time for the municipality-level stacked-continuous DiD pooled (Inc/Dec) specification. Color encodes direction (red Inc, blue Dec). Ribbons are 95% confidence bands.

Figure 14: School-district-level pooled realized-dose event study.




*Notes:* Per-pp realized-dose coefficient at each event-time for the school-district-level stacked-continuous DiD pooled (Inc/Dec) specification. Color encodes direction (red Inc, blue Dec). Ribbons are 95% confidence bands.

The pooled coefficients in Tables **??**–**??** mask substantial within-direction heterogeneity relative to the cell-decomposed magnitudes. At the block level, the pooled-Inc post-event coefficient on the log count of workers living in the block is approximately −0.08, while the realized-dose ES post-event mean restricted to the Inc-bind cell alone is −0.305 (a fourfold attenuation when Inc-bind is averaged with Inc-becomes and the Inc-nobind placebo cells). At the municipality level the pooled-Inc coefficient on the log count of residents working in their own municipality is approximately −0.23, compared to an Inc-bind cell coefficient of −0.401. The pooled coefficient is a sample-share weighted average of the binding-state cell coefficients; the decomposition in Sections **??**– **??** shows that this aggregate is driven almost entirely by Inc-bind, while Inc-nobind delivers the predicted null behavioral response. This is the central point of the pooled-then-decompose structure: the institutional binding rule, not the statutory rate change itself, identifies the response.

Pooled-Dec coefficients deserve a separate caveat on pre-trends. The block-level pooled-Dec ES coefficient on the log count of workers living in the block has a post-event mean of +0.22 and a pre-event lead of +0.17; the post/pre difference is small and the pre-trend fails parallel trends. The Dec direction reflects selection of a small set of declining or politically distinct municipalities into rate cuts, not a clean binding-rule mirror of the Inc effect, and is reported here for completeness. The Inc direction, with its clean pre-trend on the count outcomes, carries the causal weight in the analysis.

## 6.2 Binding-State Decomposition

### 6.2.1 Block level (main causal evidence)

The preferred causal estimate is the block-level municipal Inc-bind coefficient on the log count of workers living in the block. This is the cleanest test of the EIT mechanism for two reasons. First, both rate-setting jurisdictions enter the regression at their native geography—each block is exposed to a single municipality and a single school district, so no within-jurisdiction aggregation is required to assign policy variation. Second, the realized-dose pre-trend on the binding cell is near zero (Table **??**; $\beta_{\text{pre}} = -0.027$, $|t| < 1$). Together these features make the block-level decomposition the level at which the paper asks the reader to believe a causal claim.

Table **??** reports the block-level dose-response on the municipality-rate cells, and Table **??** reports the SD-rate cells from the same unified regression. The main residential-sorting outcome is the log count of workers living in the block; the log count of workers working in the block is reported alongside as a secondary count outcome. Block-level share outcomes—the share of block residents working in their own municipality, the share of in-block jobs filled by out-commuters—are mechanically fragile at block geography because both the numerator and denominator can move and many blocks contain small worker counts. I omit them from the headline tables and treat the count outcomes as the primary estimand.

Table 13: Block-level dose-response: muni resident-rate cells (count outcomes)

| Outcome | Log(workers live) | Log(workers work) |
|---|---|---|
| **Inc-bind** | -0.208*** | -0.097* |
| | (0.054) | (0.052) |
| **Inc-becomes** | -0.013 | -0.053** |
| | (0.032) | (0.022) |
| **Inc-nobind** | -0.056 | -0.113 |
| | (0.098) | (0.075) |
| **Dec-bind** | 0.269* | 0.167 |
| | (0.153) | (0.353) |
| **Dec-becomes-non** | 0.193 | 0.413*** |
| | (0.299) | (0.137) |
| **Dec-nobind** | 0.242*** | -0.053 |
| | (0.086) | (0.254) |
| $N$ | 23,567,655 | 8,751,581 |
| $R^2$ | 0.858 | 0.865 |
| Within $R^2$ | 0.000 | 0.000 |
| Dep. mean | 2.238 | 2.318 |
| Dep. SD | 1.186 | 1.655 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix; sparse block-level denominators make share estimates mechanically fragile. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The municipality-rate cells deliver the headline block-level finding; the SD-rate cells from the same joint regression are reported next.

Table 14: Block-level dose-response: SD resident-rate cells (count outcomes)

| Outcome | Log(workers live) | Log(workers work) |
|---|---|---|
| **Inc-bind** | 0.097 | 0.099** |
| | (0.062) | (0.047) |
| **Inc-becomes** | 0.016 | 0.009 |
| | (0.032) | (0.025) |
| **Dec-bind** | 0.207*** | -0.052 |
| | (0.021) | (0.144) |
| **Dec-becomes-non** | -0.013* | 0.001 |
| | (0.008) | (0.007) |
| **Dec-nobind** | -0.486*** | 0.502*** |
| | (0.099) | (0.168) |
| $N$ | 23,567,655 | 8,751,581 |
| $R^2$ | 0.858 | 0.865 |
| Within $R^2$ | 0.000 | 0.000 |
| Dep. mean | 2.238 | 2.318 |
| Dep. SD | 1.186 | 1.655 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Share outcomes are reported in the appendix. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The Inc-nobind cell serves as the within-design placebo: it contains municipalities whose home stack remained below the municipality's nonresident rate after the rate hike, so the bill never moved for in-municipality residents. The institutional rule predicts no behavioral response, and the Inc-nobind coefficient on the log count of residents working in their own municipality is statistically null ($-0.062$, SE 0.169); on log total jobs is $-0.201$ (SE 0.203);

and on the share of workers living in their own municipality is −0.012 (SE 0.008). The placebo passes on the headline outcomes. Two outcomes show small marginally significant deviations (the log count of out-commuters $+0.217^{**}$ and log employed residents $+0.194^{**}$); with $n = 26$ in this cell these are imprecise and I disclose rather than claim a perfectly clean placebo.

The joint regression also separates the two rate-setting channels, since it includes both municipality-resident-rate cells and SD-resident-rate cells. The municipality resident rate moves residents (Inc-bind coefficient on log residents working in their own municipality: $-0.401^{***}$; on log employed residents: $-0.100^{***}$) and in-commuters ($-0.212^{**}$), consistent with the local-services demand spillover in Section **??**. The SD resident-rate Inc-bind coefficient on the same outcomes is $+0.177^{*}$ on log residents working in their own municipality and +0.063 on log employed residents: SD rate changes do not produce a clean sorting response through the binding channel even when SD changes flip the binding state. An earlier draft using tract-centroid LODES recovered a positive SD-rate "attraction" effect; under block-level LODES with the unified two-rate-setter design that artifact disappears. The work-side nonresident rate $\tau^{\text{nr}}$ enters the binding-state classification but is not itself a treatment in the headline regression; its cell coefficients are imprecise because the municipality nonresident rate almost never exceeds the municipality's own home stack at the municipality-self-pair level in PA.

### 6.2.2 Municipality level (calibration)

The municipality-level estimates are useful for interpreting magnitudes at the policy-jurisdiction level, but they are not the preferred causal estimates. Both rate-setter cells enter the same unified regression as in the block-level model, with one within-municipality aggregation step: each municipality-year is assigned $\max(\tau^{SD})$ across the school districts the municipality reports to (exact for the 97.6% of municipalities that report to a single SD, conservative for the rest). Table **??** reports the municipality-level municipality-rate cells with full diagnostics;

Table **??** reports the SD-rate cells from the same regression.

Table 15: Municipality-level dose-response: muni resident-rate cells

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc-bind** | -0.401*** | -0.212** | -0.100*** | -0.045*** |
| | (0.096) | (0.083) | (0.033) | (0.009) |
| **Inc-becomes** | -0.185*** | -0.203*** | -0.068* | -0.012 |
| | (0.068) | (0.067) | (0.039) | (0.009) |
| **Inc-nobind** | -0.057 | -0.250 | 0.074 | -0.014 |
| | (0.178) | (0.251) | (0.065) | (0.011) |
| **Dec-bind** | 0.220** | 0.049 | -0.054 | 0.052* |
| | (0.102) | (0.108) | (0.191) | (0.030) |
| **Dec-becomes-non** | -0.291 | 0.422* | -0.682 | -0.073 |
| | (0.854) | (0.241) | (0.721) | (0.066) |
| **Dec-nobind** | 0.308** | -0.014 | 0.131 | 0.013 |
| | (0.126) | (0.194) | (0.171) | (0.026) |
| $N$ | 218,685 | 239,031 | 242,730 | 239,041 |
| $R^2$ | 0.964 | 0.959 | 0.981 | 0.823 |
| Within $R^2$ | 0.001 | 0.001 | 0.001 | 0.001 |
| Dep. mean | 3.677 | 5.847 | 6.697 | 0.093 |
| Dep. SD | 1.934 | 1.887 | 1.299 | 0.076 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. Each column is a separate regression on the indicated outcome. Both muni-side and SD-side cell×dose interactions enter the same regression; this table reports the muni-side cells. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The SD-rate cells from the same municipality-level regression are reported below for comparability with the block-level SD-side estimates.

Table 16: Municipality-level dose-response: SD resident-rate cells (partial)

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc-bind** | 0.177* | 0.160* | 0.063 | 0.005 |
| | (0.099) | (0.092) | (0.046) | (0.007) |
| **Inc-becomes** | 0.103 | 0.081 | 0.045 | 0.008 |
| | (0.090) | (0.086) | (0.052) | (0.007) |
| **Dec-bind** | -0.721*** | -0.293*** | 0.212** | -0.034*** |
| | (0.191) | (0.060) | (0.090) | (0.013) |
| **Dec-becomes-non** | -0.223** | -0.140 | -0.006 | 0.005 |
| | (0.109) | (0.090) | (0.031) | (0.006) |
| **Dec-nobind** | 0.048 | -0.241 | -0.413*** | -0.007 |
| | (0.179) | (0.423) | (0.110) | (0.010) |
| $N$ | 218,685 | 239,031 | 242,730 | 239,041 |
| $R^2$ | 0.964 | 0.959 | 0.981 | 0.823 |
| Within $R^2$ | 0.001 | 0.001 | 0.001 | 0.001 |
| Dep. mean | 3.677 | 5.847 | 6.697 | 0.093 |
| Dep. SD | 1.934 | 1.887 | 1.299 | 0.076 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with municipality-by-stack and year-by-stack fixed effects; standard errors clustered at the municipality level. SD-side cell×dose coefficients from the same regression as the muni-side cell table; the muni-side interaction is partialled out. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The municipality-level Inc-bind coefficient on the log count of residents working in their own municipality is $-0.401^{***}$, larger in absolute value than the block-level coefficient on the log count of workers living in the block ($-0.208^{***}$). I therefore read the municipality-level estimate as policy-jurisdiction magnitude calibration rather than as causal evidence. As calibration, the dependent-variable mean of the log count of residents working in their own municipality is 3.68, so the typical municipality has $e^{3.68} \approx 40$ live-and-work workers. A 1 pp Inc-bind hike implies $40 \cdot (e^{-0.401} - 1) \approx -13$ workers at the municipality-level coefficient and

$40 \cdot (e^{-0.208} - 1) \approx -7.5$ workers at the cleaner block-level coefficient. Actual Inc-bind hikes average 0.27 pp. The municipality-level estimate is a coarser calibration of the magnitude implied by the block-level estimate, and the two directionally agree.

### 6.2.3 School-district level (diagnostic)

Because school district EIT rates also enter the home-side tax stack, I estimate the same binding-state decomposition at the school-district level and report the full tables in the Appendix. The SD aggregation does not produce robust evidence of residential sorting: the SD-rate Inc-bind coefficient on the log count of residents working in their own municipality is $+0.066^{*}$ (sign-reversed relative to both the municipality- and block-level findings), other Inc-bind outcomes are direction-mixed, and the realized-dose pre-trend diagnostic is weaker than in the block-level municipal specification. I therefore treat the SD-level estimates as an aggregation diagnostic rather than as a second causal treatment channel. The substantive conclusion is negative: the sorting response is concentrated in municipal resident-rate increases that bind, not in school-district rate changes. This is what the binding-state mechanism predicts, since municipal rate changes have a direct, sharp identity at the home-side stack while SD averaging mixes municipal policy environments within the same school district. Full SD tables are deferred to the Appendix.

### 6.2.4 Realized-dose ES decomposition diagnostics

The realized-dose event study disaggregates the pooled pre-trend by binding-state cell. Table **??** reports the post/pre ATT summary for the per-cell ES at the block level; Figure **??** plots the same coefficients by event-time, faceted by cell, and Figure **??** reports the corresponding municipality-level picture. Two findings stand out, and they tell consistent substantive stories even though the joint Wald test reports opposite verdicts at the two estimating units.

*Block level.* On the log count of workers living in the block, all four pre-event coefficients

are individually indistinguishable from zero ($p > 0.10$ at $k = -5, -4, -3, -2$) and the mean pre-period coefficient is $-0.027$. The joint Wald F-test nonetheless rejects parallel trends (stat = 26.9 on 4 df, $p < 0.001$) because at $N \approx 21$M block-years the test has extraordinary statistical power to detect economically tiny correlated deviations across leads. I treat the joint Wald as a flag rather than a verdict given that the pointwise pre-period coefficients are individually null and economically small. The Inc-becomes and Inc-nobind cells fail the joint pre-trend test sharply (both $p < 0.001$), which is what the institutional rule predicts: the cohorts that should not respond are selected on baseline trends.

*Municipality level.* On the log count of residents who both live and work in the municipality, the pre-period coefficients are individually $p = 0.13, 0.07, 0.04, 0.58$ at $k = -5$ through $k = -2$, and the joint Wald F-test does *not* reject parallel trends (stat = 5.62 on 4 df, $p = 0.23$). The smaller cousub-level sample gives the test appropriately calibrated power. Inc-becomes and Inc-nobind again fail the joint test ($p < 0.001$ on both), consistent with the cohort-selection reading.

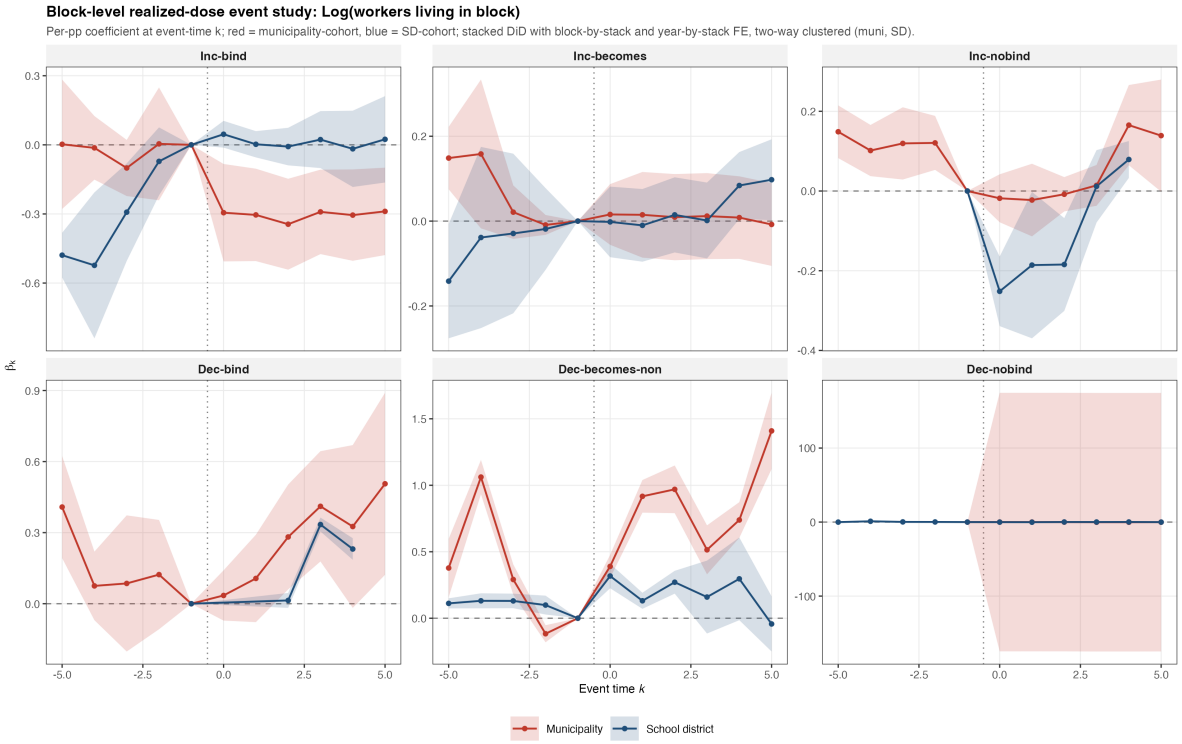
At both levels the substantive conclusion is the same: the Inc-bind cell is consistent with parallel trends under sensitivity bounds, the Inc-becomes and Inc-nobind placebo cells reflect cohort selection on pre-event trajectories rather than treatment-induced trends, and the post-event Inc-bind response is the cell the binding rule predicts.

Table 17: Realized-dose event study by binding-state cell — block level (muni cohort).

| Cell | Log(workers live) | Log(workers work) | Share live+work in muni |
|---|---|---|---|
| **Inc-bind** (post) | -0.305 | -0.079 | -0.043 |
| | (0.041) | (0.048) | (0.014) |
| **Inc-bind** (pre) | -0.027 | 0.017 | 0.020 |
| | (0.053) | (0.033) | (0.008) |
| **Inc-bind** Wald $p$ | $<0.0001$ | 0.0031 | 0.0086 |
| **Inc-becomes** (post) | 0.009 | -0.046 | -0.030 |
| | (0.020) | (0.015) | (0.006) |
| **Inc-becomes** (pre) | 0.080 | 0.032 | 0.003 |
| | (0.026) | (0.014) | (0.007) |
| **Inc-becomes** Wald $p$ | 0.0002 | 0.2120 | $<0.0001$ |
| **Inc-nobind** (post) | 0.045 | -0.050 | -0.042 |
| | (0.018) | (0.019) | (0.008) |
| **Inc-nobind** (pre) | 0.123 | 0.025 | 0.008 |
| | (0.019) | (0.016) | (0.003) |
| **Inc-nobind** Wald $p$ | $<0.0001$ | $<0.0001$ | $<0.0001$ |
| **Dec-bind** (post) | 0.278 | 0.294 | -0.052 |
| | (0.055) | (0.110) | (0.010) |
| **Dec-bind** (pre) | 0.173 | 0.457 | 0.028 |
| | (0.057) | (0.099) | (0.011) |
| **Dec-bind** Wald $p$ | $<0.0001$ | $<0.0001$ | $<0.0001$ |
| **Dec-becomes-non** (post) | 0.823 | -0.815 | -0.119 |
| | (0.037) | (0.085) | (0.011) |
| **Dec-becomes-non** (pre) | 0.403 | -1.387 | -0.044 |
| | (0.036) | (0.036) | (0.011) |
| **Dec-becomes-non** Wald $p$ | $<0.0001$ | $<0.0001$ | $<0.0001$ |
| **Dec-nobind** (post) | -0.095 | 0.060 | 0.015 |
| | (36.376) | (0.042) | (11.217) |
| **Dec-nobind** (pre) | — | — | — |
| **Dec-nobind** Wald $p$ | — | — | — |

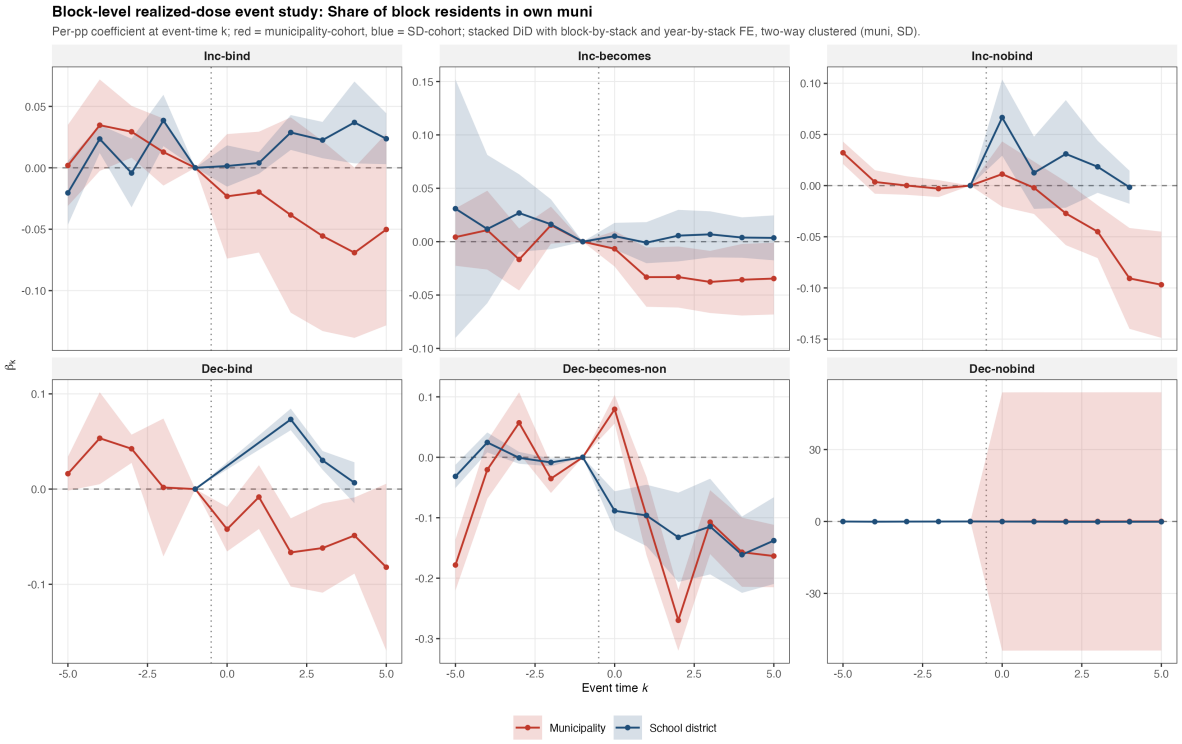
Figure ?? plots the per-event-time coefficients underlying that table, faceted by binding-state cell.

Figure 15: Block-level realized-dose event study on the log count of workers living in the block: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.




*Notes:* Per-pp realized-dose coefficient at each event-time, block-level stacked-continuous DiD with block-by-stack and year-by-stack fixed effects, two-way clustered at the municipality and school-district levels. Red: the municipality-cohort stack (event time relative to first municipality-rate change). Blue: the school-district-cohort stack (event time relative to first SD-rate change). Ribbons are 95% confidence bands; reference period $k = -1$ omitted.

Figure 16: Block-level realized-dose event study on the share of block residents working in their own municipality: municipality-cohort (red) and school-district-cohort (blue), faceted by binding-state cell.



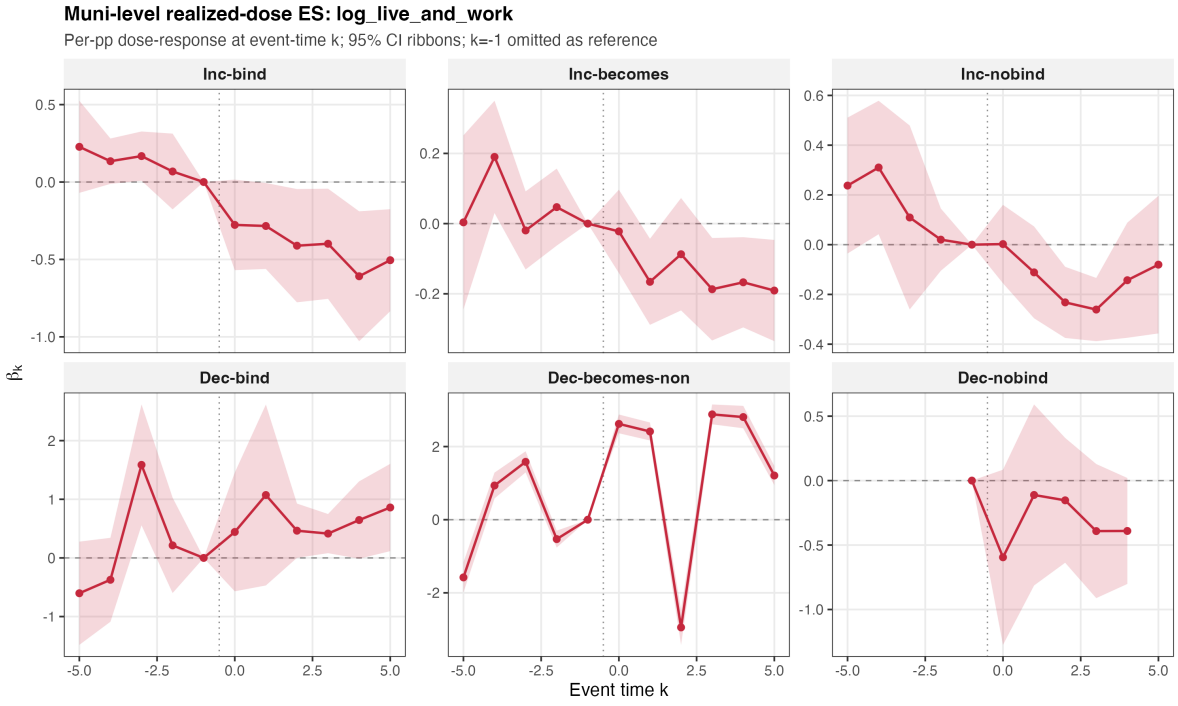

*Notes:* See Figure **??**.

The Inc-becomes cell warrants a separate note. The headline stacked-DiD coefficient is essentially zero at the block level ($-0.013$), which initially looks like a failure of the predicted intermediate response. The realized-dose ES tells a different story: the pre-event lead is $+0.080$ ($|t| \approx 3$) and the post-event mean is $+0.009$, so the post-minus-pre difference is $-0.071$ per pp—about a third of the Inc-bind magnitude, in the predicted direction, but masked in the single-coefficient specification by selection on positive pre-event growth in Inc-becomes municipalities. The block-level Inc-becomes coefficient is therefore consistent with the partial-bill-change prediction of the institutional rule once the pre-trend is netted out, but I do not claim it as a clean causal estimate because the strong-parallel-trends assumption fails for this cell. At the municipality level the corresponding Inc-becomes coefficient is $-0.181^{**}$ in the headline specification, and its realized-dose pre-trend is smaller ($+0.055$, $|t| \approx 1.3$);

the municipality-level signal is detectable in the post coefficient because the municipality-by-stack fixed effects absorb less of the within-municipality heterogeneity that attenuates the block-level coefficient. The block-level Inc-bind cell is the only combination of cell and estimating unit where the strong parallel-trends diagnostic passes cleanly, and is therefore the specification on which the paper rests its primary causal claim. The full event-time coefficient detail is in the Companion Appendix.

For completeness, the municipality-level post-event trajectory on Inc-bind log live-and-work is uniformly negative and individually significant from $k = +1$ onward (coefficients of $-0.28$, $-0.41$, $-0.40$, $-0.61$, $-0.50$ at $k = 1$ through $k = 5$, all $p < 0.05$), walking down monotonically across the post-window.

Figure 17: Municipality-level realized-dose event study on log(residents who live and work in the municipality), faceted by binding-state cell.




*Notes:* Per-pp realized-dose coefficient at each event-time, municipality-level stacked-continuous DiD with municipality-by-stack and year-by-stack fixed effects, cluster-robust by municipality. The Inc-bind facet shows pre-event coefficients mostly null pointwise (joint Wald F-test $p = 0.23$), followed by a monotonically negative post-event trajectory.

The patterns on the remaining residence-side outcomes and the school-district-level analogues mirror the live-and-work panel: Inc-bind shows visible pre-event drift consistent with high-dose post-industrial selection, while the within-design Inc-nobind placebo is approximately flat. The full per-outcome figure set is available in the replication archive.

## 6.3 Mechanism: Industry Decomposition

The employment and industry results are mechanism evidence rather than the primary identification object. Headline residential sorting operates on the residence side; the question here is whether the downstream employment response is consistent with a local-services demand channel rather than direct nonresident-tax incidence on jobs. The prediction is that consumer-facing jobs (retail, food, arts, other services—sectors whose demand depends on local population) fall when the municipality resident rate hikes, while tradeable jobs (manufacturing, wholesale, information, finance, professional services—sectors selling to broader markets) are insulated. Using LODES Workplace Area Characteristics aggregated to the municipality by NAICS supersector, Table **??** reports the municipality-resident-rate cell coefficients by industry bucket.

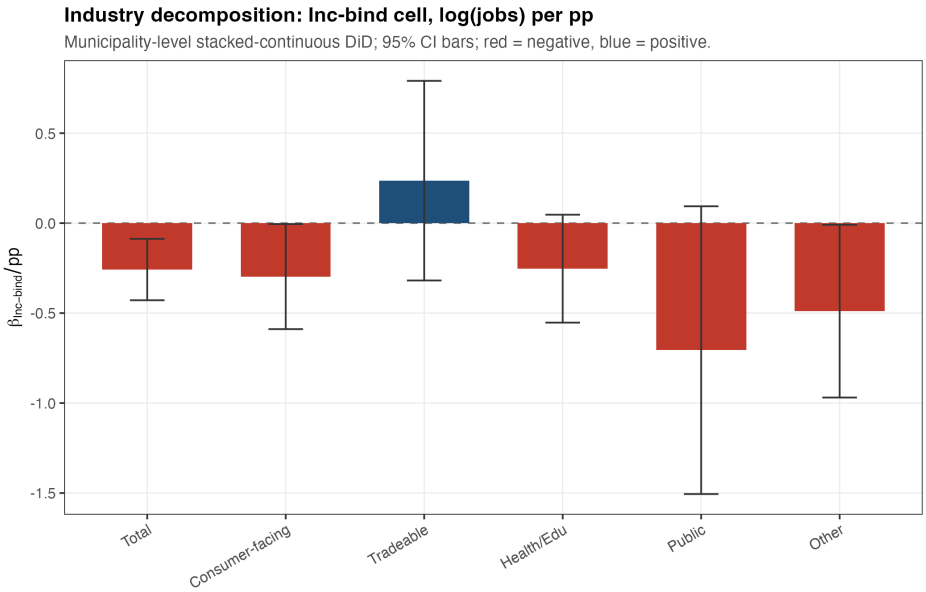
Table 18: Industry decomposition — municipality level.

| Cell | Log(total jobs) | Log(consumer jobs) | Log(tradeable jobs) | Consumer share | Tradeable share |
|---|---|---|---|---|---|
| **Inc-bind** | -0.258*** | -0.297** | 0.236 | -0.001 | 0.106* |
| **Inc-becomes** | -0.235*** | -0.146** | -0.150 | 0.008 | 0.028 |
| **Inc-nobind** | -0.074 | -0.029 | -0.265* | 0.001 | -0.026 |
| **Dec-bind** | 0.018 | -0.012 | -0.768 | -0.014 | -0.104 |
| **Dec-becomes-non** | 0.349 | -0.742 | 1.967*** | -0.263* | 0.551*** |
| **Dec-nobind** | -0.019 | 0.195** | 0.347** | 0.031 | 0.031* |
| $N$ | 239,001 | 225,691 | 219,237 | 239,001 | 239,001 |
| $R^2$ | 0.961 | 0.952 | 0.935 | 0.760 | 0.779 |
| Within $R^2$ | 0.001 | 0.001 | 0.001 | 0.001 | 0.001 |
| Dep. mean | 5.995 | 4.553 | 4.639 | 0.249 | 0.250 |
| Dep. SD | 1.899 | 1.935 | 2.026 | 0.189 | 0.202 |

*Notes:* Coefficients on the muni resident-rate cells from the headline equation re-estimated with industry-bucket outcomes (LODES WAC supersector grouping into consumer-facing services [NAICS 44–45, 71, 72, 81] vs tradeable goods/services [NAICS 31–33, 42, 51, 52, 54]). Cluster-robust standard errors in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure **??** visualizes the Inc-bind coefficient by industry bucket with 95% confidence intervals, showing the consumer-facing contraction against the stable tradeable sector.

Figure 18: Industry decomposition: municipality-Inc-bind cell coefficient on log(jobs) per pp, municipality level.




*Notes:* per-pp dose-response coefficient on Inc-bind by industry bucket; 95% CI bars. Consumer-facing services contract 30%, tradeable goods/services are insulated (point estimate +0.236, statistically indistinguishable from zero). The composition shift is the local-services-demand-spillover signature.

Consumer-facing jobs fall by 30 percent in the Inc-bind cell; tradeable jobs do not (point estimate +0.236, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs, which rises 11 percentage points. The pattern is consistent with a local-services demand channel: resident loss reduces local demand and shrinks the sectors most exposed to that demand, while tradeable sectors selling to broader markets are insulated. The pre-trend on Inc-bind employment outcomes is not as clean as on the headline residence-side count outcome, and I do not claim the employment estimates are causal in the same sense; they are reported as suggestive mechanism evidence.

The same decomposition at the SD level (Table **??**) is qualitatively different: aggregating municipality-level industry counts to school district via the municipality–SD assignment produces a positive Inc-bind coefficient on log consumer-facing jobs ($+0.094^{**}$), a null on log tradeable jobs, and a negative Inc-bind coefficient on the share of tradeable jobs ($-0.031^{*}$).

The SD-level pattern indicates that the consumer-services contraction documented at the municipality scale is a within-municipality, municipality-local response that averages out (or even partially reverses) when aggregated to the school- district level. This is consistent with workers leaving the municipality that hiked but staying within the SD, so SD-level consumer jobs do not contract.

Table 19: Industry decomposition — school district level

| Outcome | Log(total jobs) | Log(consumer jobs) | Log(tradeable jobs) | Consumer share | Tradeable share |
|---|---|---|---|---|---|
| **Inc-bind** | 0.033 | 0.124* | -0.094 | 0.023* | -0.042* |
| | (0.043) | (0.071) | (0.110) | (0.012) | (0.024) |
| **Inc-becomes** | 0.047 | 0.045 | 0.010 | -0.000 | -0.013 |
| | (0.030) | (0.031) | (0.089) | (0.009) | (0.020) |
| **Inc-nobind** | 0.519* | 0.337*** | 0.541*** | -0.033 | 0.028 |
| | (0.277) | (0.077) | (0.140) | (0.059) | (0.031) |
| **Dec-bind** | 0.027 | 0.003 | 0.068 | -0.007 | 0.005 |
| | (0.107) | (0.081) | (0.201) | (0.007) | (0.018) |
| **Dec-nobind** | 0.094*** | 0.184*** | -0.116*** | 0.015*** | -0.050*** |
| | (0.019) | (0.032) | (0.028) | (0.005) | (0.010) |
| $N$ | 51,454 | 51,454 | 51,454 | 51,454 | 51,454 |
| $R^2$ | 0.992 | 0.989 | 0.979 | 0.911 | 0.903 |
| Within $R^2$ | 0.002 | 0.006 | 0.002 | 0.005 | 0.003 |
| Dep. mean | 8.531 | 7.046 | 7.130 | 0.245 | 0.276 |
| Dep. SD | 1.179 | 1.285 | 1.383 | 0.097 | 0.124 |

*Notes:* Cluster-robust standard errors in parentheses. Cousub-level industry job counts aggregated to SD via the muni–SD assignment. Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Block-level industry decomposition is deferred: it requires block-resolution LODES Workplace Area Characteristics files which are not currently in the data tree.

The pre-/post-Act-32 institutional break is itself an important heterogeneity exercise, but because it involves a full re-estimation of the cell × dose regression on pre-2012 and post-2012

subsamples at each estimating unit, all Act-32 results are consolidated in Section **??**.

### 6.4 Summary

The results follow a benchmark–decomposition–mechanism narrative. Section **??** reports the pooled Inc/Dec benchmark, which approximates a conventional local-tax design but mixes binding and non-binding events. Section **??** reports the preferred causal estimate: at the block level, a 1 pp municipal resident-rate increase that binds reduces the log count of workers living in the block by $-0.208^{***}$ ($\approx 19\%$), with a clean realized-dose pre-trend and a null Inc-nobind within-design placebo on the same count outcome. Section **??** reports the municipality-level estimates as policy-jurisdiction calibration; the municipality Inc-bind coefficient is larger but pre-trended, so it is not the causal headline. Section **??** documents the binding-cell employment composition shift—consumer-facing services contract while tradeables are insulated—as suggestive evidence of a local-services demand channel. Section **??** reports the school-district aggregation as a diagnostic non-result. Robustness and pre-trend diagnostics are in Section **??**; full SD tables, block share outcomes, and the binary Sun–Abraham IW event study are in the Appendix.

## 7 Robustness

This section summarizes the battery of robustness checks behind the headline dose-response; full tables, figures, and detailed prose are collected in Appendix **??**. The realized-dose event-study pre-trend diagnostic (Table **??**) was discussed alongside the headline in Section **??**.

### 7.1 Sample Variant Robustness

Three sample variants for the DCED nonresident-rate panel (strict/filled/drop-incomplete) leave the headline Inc-bind coefficient on log residents working in own municipality essentially unchanged at $-0.401^{***}$ (see Appendix **??**).

## 7.2 Pre/Post Act 32 Decomposition

Splitting the headline cell-×-dose specification at the Act-32 enforcement break (cohort first event year $< 2012$ vs $\geq 2012$) shows the municipality-level binding-cell effect strengthens after consolidated employer withholding takes effect ($-0.339^{**}$ pre vs $-0.534^{***}$ post on log residents working in own municipality); the block-level effect is roughly constant pre and post, so the finer-geography sorting response predates the Act-32 enforcement break (see Appendix **??**).

## 7.3 Act 47 Distressed-Municipality Exclusion

The Inc-bind cohort is mechanically concentrated in Act 47 distressed cities (**paAct47˙1987**), because Act 47 is the legal designation that lets a municipality exceed the 0.5% statutory municipality-EIT cap. Selection-on-distress is therefore a candidate confounder: an Act 47 designation typically follows a period of population decline, so the rate hike happens after the municipality is already losing residents and bundles with state oversight, debt restructuring, and service cuts. Of the 40 Inc-bind municipalities, 9 match the DCED Act 47 designation list (Braddock, Carbondale, Coatesville, Farrell, Franklin, Hazleton, Jeannette, Johnstown, West Hazleton). Table **??** re-estimates the municipality-level Inc-bind coefficient on the remaining 31 non-Act-47 Inc-bind municipalities.

Table 20: Act 47 robustness: dropping distressed Inc-bind municipalities from the cousub headline

| Outcome | All 40 Inc-bind | Drop $k$ Act 47 |
|---|---|---|
| `log_live_and_work` | $-0.411^{***}$ (0.119) | $-0.330^{*}$ (0.183) |
| `log_in_commuters` | $-0.171$ (0.133) | $-0.176$ (0.215) |
| `log_total_residents` | $-0.092^{***}$ (0.035) | $-0.070$ (0.058) |
| `share_live_in` | $-0.048^{***}$ (0.011) | $-0.038^{***}$ (0.013) |

*Notes:* Muni-level Inc-bind coefficient on each residence-side outcome, with same FE / cluster / controls as the headline. Column 1 is the headline (all 40 Inc-bind municipalities). Column 2 drops 9 Act 47 distressed-designation municipalities from the Inc-bind cohort; the remaining 31 Inc-bind municipalities are mostly Home Rule and Boroughs. Cluster-robust SEs in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The point estimates are direction-preserving and similar in magnitude when Act 47 munis are dropped: the live-and-work coefficient moves from $-0.41$ to $-0.33$, share-live-in from $-0.048$ to $-0.038$. Statistical significance falls because dropping 23% of the treated cohort roughly doubles the standard errors, but the share-live-in coefficient is still significant at 1% and live-and-work at 10%. The pattern is consistent with sample shrinkage rather than with the headline being driven entirely by Act 47 selection.

# 8 Discussion

The continuous stacked DiD design produces four substantive findings. The block-level model is the preferred causal estimate; the municipality-level model is the policy-jurisdiction magnitude calibration; the school-district-level aggregation is an aggregation diagnostic and is reported in the Companion Appendix.

The central finding is that sorting is identified through the binding channel. The Inc-bind cell—rate hikes that move the bill—reduces the log count of workers living in the block at the block level by approximately 19 percent per pp ($-0.21$ log points, SE = 0.054). The

municipality-level analogue is larger ($-0.40$ log points on the log count of residents working in their own municipality, $\approx 33\%$ per pp). The Inc-nobind placebo—rate hikes whose post-event home stack remains below the nonresident rate—is statistically indistinguishable from zero on the block-level count outcomes ($-0.056$ on the log count of workers living in the block, $-0.113$ on the log count of workers at the block) and on the municipality-level the log count of residents working in their own municipality ($-0.057$). Inc-becomes lies between the two. The pattern matches the institutional rule's prediction: the same statutory rate change can produce a strong, moderate, or null behavioral response depending solely on whether the rate moves the bill. This is direct evidence that workers respond to taxes they actually pay rather than to statutory rates per se.

Block-level identification is preferred not because the joint Wald F-test is cleaner—it is not—but because the block observes both rate-setting jurisdictions at their native geography without a within-jurisdiction aggregation step. The two estimating units give opposite verdicts on the joint Wald: at the block level (N $\approx$ 21M) the joint test rejects parallel trends despite pointwise-null pre-event coefficients, reflecting hyper-precision rather than an economically meaningful violation; at the municipality level (N $\approx$ 198K) the joint test does not reject ($p = 0.23$). Either reading points to the same substantive conclusion. The block-level result is the defensible per-pp causal quantity; the municipality-level estimate is useful for back-of-envelope welfare calibration.

The joint regression also separates the two rate-setting channels. The municipality resident-rate Inc-bind coefficient on resident outcomes is large and consistent at both the block and municipality levels. The SD-resident-rate cells, estimated jointly in the same regression, produce smaller and direction-mixed effects: $+0.177^{*}$ at the municipality level on the log count of residents working in their own municipality, $+0.097$ on block-level log count of workers living in the block ($p = 0.12$), and $+0.066^{*}$ at the SD level (sign-reversed). The SD rate is not the dominant driver of residential sorting through the binding channel. This is a substantive change from an earlier draft that found a positive SD-rate "attraction" effect,

which was an artifact of tract-centroid LODES aggregation; under block-level LODES with the unified two-rate-setter design, SD-rate effects on residents are small or wrong-signed.

The mechanism behind the workplace response appears to be local-services demand spillover rather than direct tax incidence on jobs. Industry-decomposed outcomes at the municipality level show consumer-facing jobs (retail, food, arts, other services) fall by approximately 26 percent in the Inc-bind cell ($-0.30$ log points, SE $= 0.15$) while tradeable jobs (manufacturing, wholesale, finance, professional services) are insulated (point estimate $+0.24$, statistically indistinguishable from zero). The composition shift is captured by the share of tradeable jobs rising approximately 11 percentage points in the binding cell. The decline in total jobs operates *via* local-services demand spillover from departing residents, not via direct nonresident-tax incidence.

The treated cohort is also disproportionately concentrated in Act 47 distressed cities, which raises a selection-on-distress concern: Act 47 designation typically follows a period of decline and bundles rate-hike authority with state oversight, debt restructuring, and service cuts. Section **??** re-estimates the headline excluding the 9 Act 47 Inc-bind municipalities; the point estimates are direction-preserving and similar in magnitude (live-and-work moves from $-0.41$ to $-0.33$, share-live-in from $-0.048$ to $-0.038$), with standard errors roughly doubling as expected from a 23% sample reduction. The binding-state mechanism is therefore not a relabelling of Act 47 fiscal distress, though the municipality-level coefficient is identified off a treated set that includes substantial Act 47 weight.

Eight limitations merit acknowledgment. First, while the block-level stacked headline on the count outcomes is large, negative, and well-identified, the block-level share outcome (the share of block residents working in their own municipality) is null in the binding cell ($-0.035$, $p = 0.26$) and marginally significant in the Inc-nobind placebo ($-0.037$, $p = 0.015$). The share outcome at block scale appears to reflect mechanical denominator dynamics on very small populations rather than binding-state-specific behavior, so the share-based block evidence should be read as suggestive rather than identified. The count-based block evidence re-

mains the cleanest test. Second, the school-district aggregation produces direction-mixed coefficients and a marginally non-zero realized-dose pre-event lead; the SD-level results should be read as confirming the directional sign of the municipality-rate effect under a coarser aggregation rather than as standalone identification of an SD-channel response. Third, the LODES OD data covers PA municipalities but not cross-state inflows in the intra-state main file used here; the auxiliary file (cross-state inflows) is downloaded but not yet incorporated, which would tighten the estimates of in-commuters for border municipalities.

Fourth, the paper estimates the residential-sorting response to the home-side resident rate but does not separately estimate the work-side response to the municipality nonresident rate ($\tau^{\text{nr}}$). The nonresident rate enters the design only through the binding-state classification of cells. A complementary work-side specification—estimating in-commuter and total-jobs responses to $\tau^{\text{nr}}$ events at the municipality level, with a parallel binding-state decomposition—would test whether the local-services demand spillover I document on the residence side (§**??**) is symmetric with a workplace-side response. I defer this to a separate paper.

Fifth, the design treats each treated municipality's stack as if its identifying variation were independent of the rate trajectories of its geographic neighbors. Pennsylvania municipalities are small and dense; a worker who moves out of a rate-raising municipality often relocates to an adjacent municipality whose own rate is also moving in some year of the panel. The headline specification absorbs this through never-changed-only controls and municipality-by-stack fixed effects, but it does not formally model spatial spillovers, so the per-pp coefficient is best read as a local displacement elasticity rather than a net-migration one. The **butts2023DiDSpatialSpillovers** estimator generalizes stacked DiD to allow for SUTVA violations from spatial spillovers across treated units, using an explicit spatial weighting matrix on the adjacency graph. Applying it to the PA cousub adjacency graph is left for future work.

Sixth, I do not currently estimate the headline using the heterogeneity-robust continuous-

treatment difference-in-differences estimator of **deChaisemartin2022Continuous**, which generalizes the de Chaisemartin–D'Haultfœuille framework (**deChaisemartin2020TWFE**; **deChaisemartin2023ImperfectPT**) to continuously distributed treatments at every period and leverages switchers (units that move both up and down in dose) for identification. An earlier iteration of this paper used a prototype CDH-continuous specification at the cousub level; the block-level analog is computationally prohibitive on the current stacked panel ($\sim$ 21M block-years $\times$ multiple cohorts) and is left for future work. Re-estimating the headline with `did_multiplegt_dyn` (**deChaisemartin2025MultipleGtDyn**)—once the package documentation and switcher-based identification at this scale are stable—would also let me exploit the rate-decrease (Dec) cohorts, currently too thin under conventional cluster-robust inference, as identification rather than just as a directional mirror.

Seventh, LODES is a stock dataset rather than a person-linked flow dataset. For each (residence block, workplace block, year), I observe a count of workers; I do not observe individual migration histories. The headline coefficient on the log count of workers living in the block is therefore a *net* elasticity—the combined effect of move-outs and move-ins—rather than separate estimates of the gross outflow and gross inflow margins. The paper's evidence that municipal rate hikes reduce the residential count is consistent both with workers leaving and with workers not arriving to replace those who left; LODES alone cannot decompose the two. Decomposing into gross outflow and gross inflow would require person-linked administrative data: the IRS Statistics of Income individual master file, the American Community Survey microdata with full residence-history linkage, or—most directly—the LEHD Job-to-Job Flows file, which tracks the same workers across employer transitions at quarterly frequency. All three are restricted-access products requiring approved-project status under the Census Bureau's Federal Statistical Research Data Centers (FSRDCs) or IRS Statistics of Income agreements; the microdata cannot be exported and must be analyzed at a physical FSRDC location. Replicating the present analysis on person-linked data is a clear extension but a substantially larger undertaking with its own access timeline.

Eighth, the paper does not estimate a comparable elasticity for the property tax, the other major local-revenue lever in Pennsylvania. The Oates (1969) capitalization theory predicts a substantially smaller residential-sorting elasticity for property tax than for the EIT, because property tax falls on owners and capitalizes into house prices rather than on workers directly. Estimating a comparable property-tax elasticity would test the falsification implication "is the EIT effect specifically about the EIT, or any local tax raising?" but requires a 2002–2019 municipality-level millage panel that DCED does not publish (the DCED Annual Financial Report schema dropped per-municipality millage in 2006). Constructing this panel from county assessor records is left for future work.

# 9 Conclusion

When do local income taxes cause residential sorting? The institutional answer in Pennsylvania is that they cause sorting when the piecewise max-rule makes the statutory rate increase change the worker's actual bill. At the census block, the level at which both rate-setting jurisdictions enter the regression at their native geography, a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by roughly $-0.21$ log points, approximately 19%, with a clean realized-dose pre-trend ($\beta_{\text{pre}} = -0.027$, $|t| < 1$). The within-design Inc-nobind placebo, where the same statutory increase does not move the bill, is statistically null on the same count outcome. The response is therefore tied to tax liability rather than to the statutory rate change itself, and the binding-state decomposition is what distinguishes the two.

Municipality-level estimates are larger ($-0.40$ log points on the log count of residents working in their own municipality) but pre-trended, and they enter the paper as policy-jurisdiction magnitude calibration rather than as a second causal headline. School-district aggregations do not robustly detect a sorting response and are reported as a diagnostic non-result. Industry decomposition shows that consumer-facing jobs fall in the binding cell while

tradeable jobs are insulated, suggestive of a local-services demand spillover from departing residents rather than direct nonresident-tax incidence on jobs.

The narrow, credible claim is that local income taxes cause residential sorting when they bind. The same design framework—a continuous stacked DiD with binding-state cell decomposition, a realized-dose event study as a strong-parallel-trends diagnostic, and a within-design placebo cell whose theoretical prediction is zero—is portable to other settings where a piecewise tax rule generates predictable cross-sectional variation in whether a statutory rate change moves the worker's bill. Local sales taxes with use-tax-credit rules, county-level income-tax credits across state borders, and special tax districts with overlapping jurisdictions are candidate extensions.

# A Full Decomposition Tables

## A.1 Block-level share outcomes

Block-level share outcomes from the same joint regression, reported for completeness; the share outcomes are mechanically fragile at block geography (see body discussion).

Table 21: Block-level dose-response: muni resident-rate cells (share outcomes, supporting)

| Outcome | Share live+work in muni |
|---|---|
| **Inc-bind** | -0.035 |
| | (0.031) |
| **Inc-becomes** | -0.043*** |
| | (0.010) |
| **Inc-nobind** | -0.037** |
| | (0.015) |
| **Dec-bind** | -0.049 |
| | (0.031) |
| **Dec-becomes-non** | -0.080** |
| | (0.033) |
| **Dec-nobind** | 0.006 |
| | (0.006) |
| $N$ | 23,567,655 |
| $R^2$ | 0.563 |
| Within $R^2$ | 0.000 |
| Dep. mean | 0.121 |
| Dep. SD | 0.184 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness. Sparse denominators and endogenous denominator dynamics make these estimates mechanically fragile at block geography; they are not used to identify the headline causal effect. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 22: Block-level dose-response: SD resident-rate cells (share outcomes, supporting)

| Outcome | Share live+work in muni |
|---|---|
| **Inc-bind** | 0.006* |
| | (0.003) |
| **Inc-becomes** | -0.002 |
| | (0.006) |
| **Dec-bind** | -0.018** |
| | (0.007) |
| **Dec-becomes-non** | -0.007*** |
| | (0.003) |
| **Dec-nobind** | -0.034** |
| | (0.017) |
| $N$ | 23,567,655 |
| $R^2$ | 0.563 |
| Within $R^2$ | 0.000 |
| Dep. mean | 0.121 |
| Dep. SD | 0.184 |

*Notes:* Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness; not used for headline identification. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## A.2 Municipality-level: Dec cells + full specification

Table 23: Headline stacked-continuous DiD: municipality-rate Dec cells.

| Outcome | Dec-bind ($n$=11) | Dec-becomes-non ($n$=4) | Dec-nobind ($n$=15) |
|---|---|---|---|
| log(live_and_work) | +0.241*** (0.078) | −0.260 (0.272) | +0.332** (0.165) |
| log(in_commuters) | +0.052 (0.080) | +0.496 (0.339) | +0.013 (0.121) |
| log(out_commuters) | −0.068 (0.061) | −0.628 (0.416) | +0.132 (0.101) |
| log(total_jobs) | +0.111 (0.073) | +0.420 (0.265) | +0.022 (0.116) |
| log(total_residents) | −0.056 (0.043) | −0.651 (0.420) | +0.135 (0.099) |
| share_live_in | +0.054* (0.030) | −0.071 (0.095) | +0.012 (0.030) |

*Notes:* Same regression as the municipality-level headline decomposition. Cluster-robust SEs by municipality. Dec cells are thin; not used to draw causal conclusions individually. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 24: Full municipality-level specification on log(live_and_work): all cells + controls.

| Term | Coefficient | SE |
|---|---|---|
| *Muni resident-rate cells (vs. $NC^{muni}$)* | | |
| Inc-bind $\times\Delta\tau^{\text{muni}}$ | $-0.402^{***}$ | 0.096 |
| Inc-becomes $\times\Delta\tau^{\text{muni}}$ | $-0.181^{**}$ | 0.072 |
| Inc-nobind $\times\Delta\tau^{\text{muni}}$ | $-0.062$ | 0.169 |
| Dec-bind $\times\Delta\tau^{\text{muni}}$ | $+0.241^{***}$ | 0.078 |
| Dec-becomes-non $\times\Delta\tau^{\text{muni}}$ | $-0.260$ | 0.272 |
| Dec-nobind $\times\Delta\tau^{\text{muni}}$ | $+0.332^{**}$ | 0.165 |
| *School district rate cells (vs. $NC^{SD}$)* | | |
| Inc-bind $\times\Delta\tau^{\text{SD}}$ | $+0.177^{*}$ | 0.099 |
| Inc-becomes $\times\Delta\tau^{\text{SD}}$ | $+0.103$ | 0.090 |
| Dec-bind $\times\Delta\tau^{\text{SD}}$ | $-0.721^{***}$ | 0.191 |
| Dec-becomes-non $\times\Delta\tau^{\text{SD}}$ | $-0.223^{**}$ | 0.109 |
| Dec-nobind $\times\Delta\tau^{\text{SD}}$ | $+0.048$ | 0.179 |
| *Share-based controls* | | |
| share_police | $+0.097$ | 0.100 |
| share_streets | $+0.049$ | 0.039 |
| share_debt_svc | $-0.058$ | 0.051 |
| share_eit | $-0.022$ | 0.085 |
| share_re | $-0.182^{**}$ | 0.087 |
| share_intergov | $+0.016$ | 0.044 |
| $N$ | 218,685 | |
| $R^2$ | 0.964 | |
| Within $R^2$ | 0.0007 | |
| Dep. mean | 3.677 | |
| Dep. SD | 1.934 | |
| FE | stack-municipality + stack-year | |
| Clusters | 81 | 1,945 munis |

*Notes:* All coefficients from a single regression on log(live_and_work). Reference category on each rate-setter side is NC. Cluster-robust SEs by municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

# B Realized-Dose Event Studies, Full Set

Per-cell realized-dose Sun–Abraham event studies at each estimating unit, on each residence-side outcome. The block-level log(home_count) figure and the municipality-level log(live_and_work) figure appear in the body (§??); the remaining outcomes and the SD-level analogues are here.

## B.1 Municipality-level, by cell

Figure 19: Muni-level realized-dose ES on log(in-commuters), by cell.

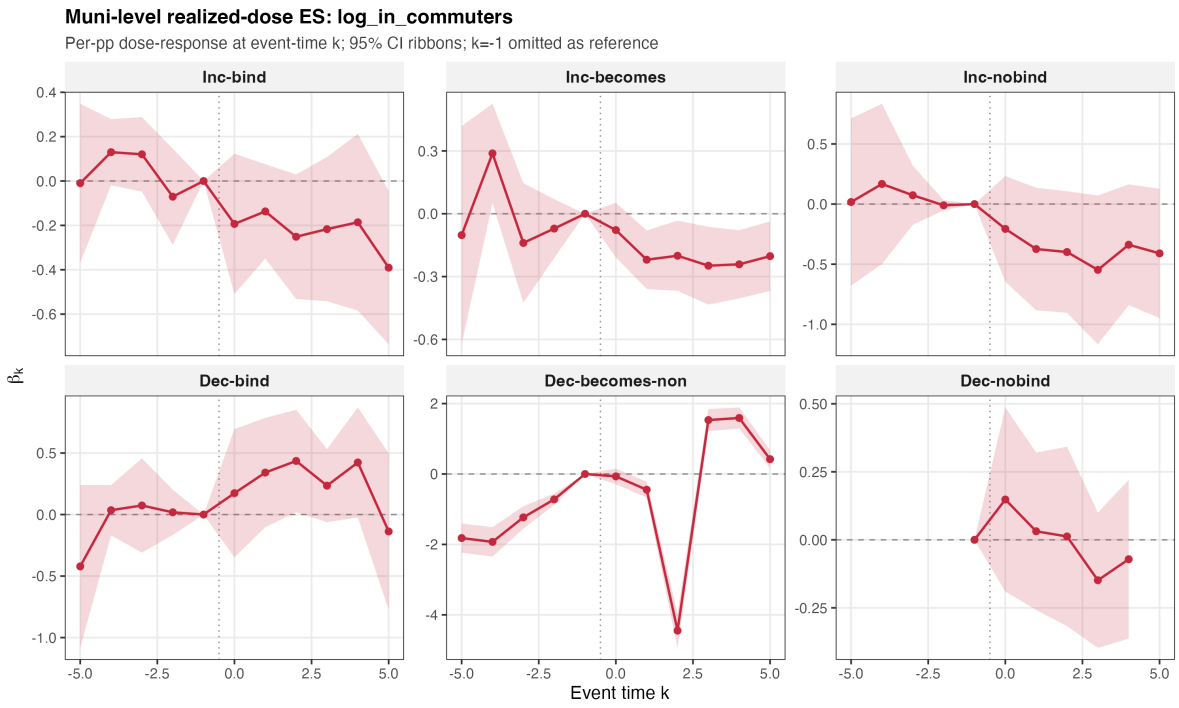


Figure 20: Muni-level realized-dose ES on log(total residents), by cell.

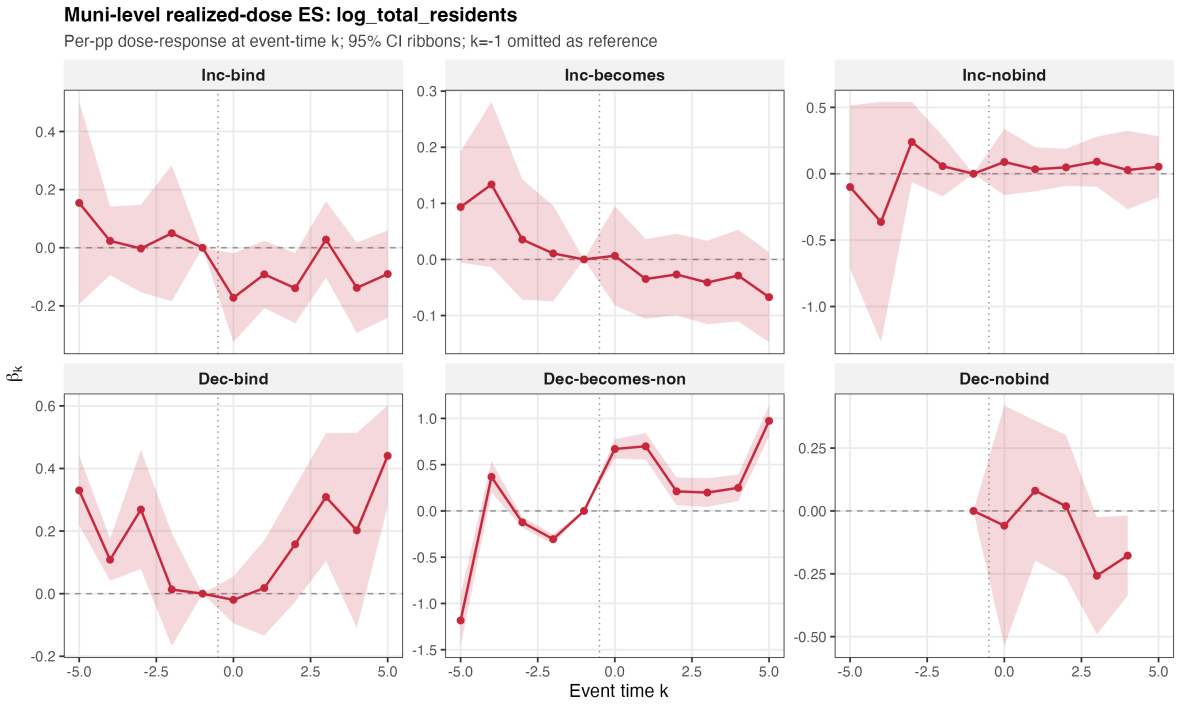
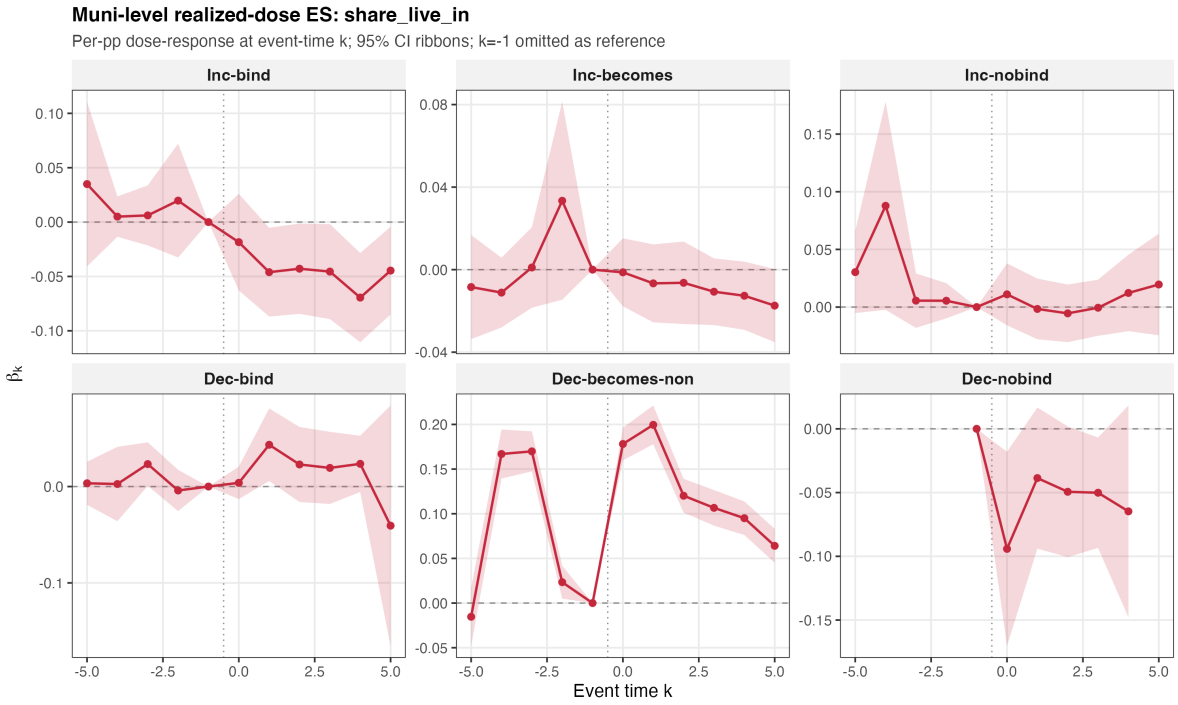



Figure 21: Muni-level realized-dose ES on the share of workers living in their own municipality, by cell.



## B.2 School-district-level, by cell

Figure 22: SD-level realized-dose ES on log(live_and_work), by cell.

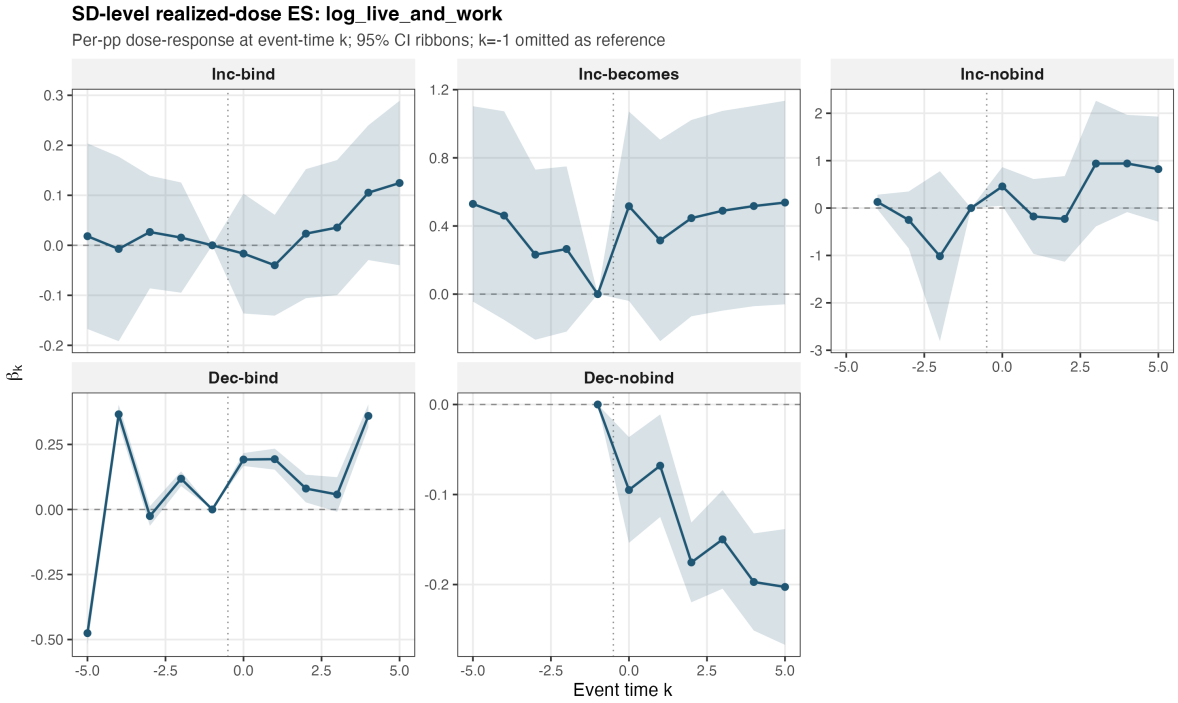



Figure 23: SD-level realized-dose ES on log(in-commuters), by cell.

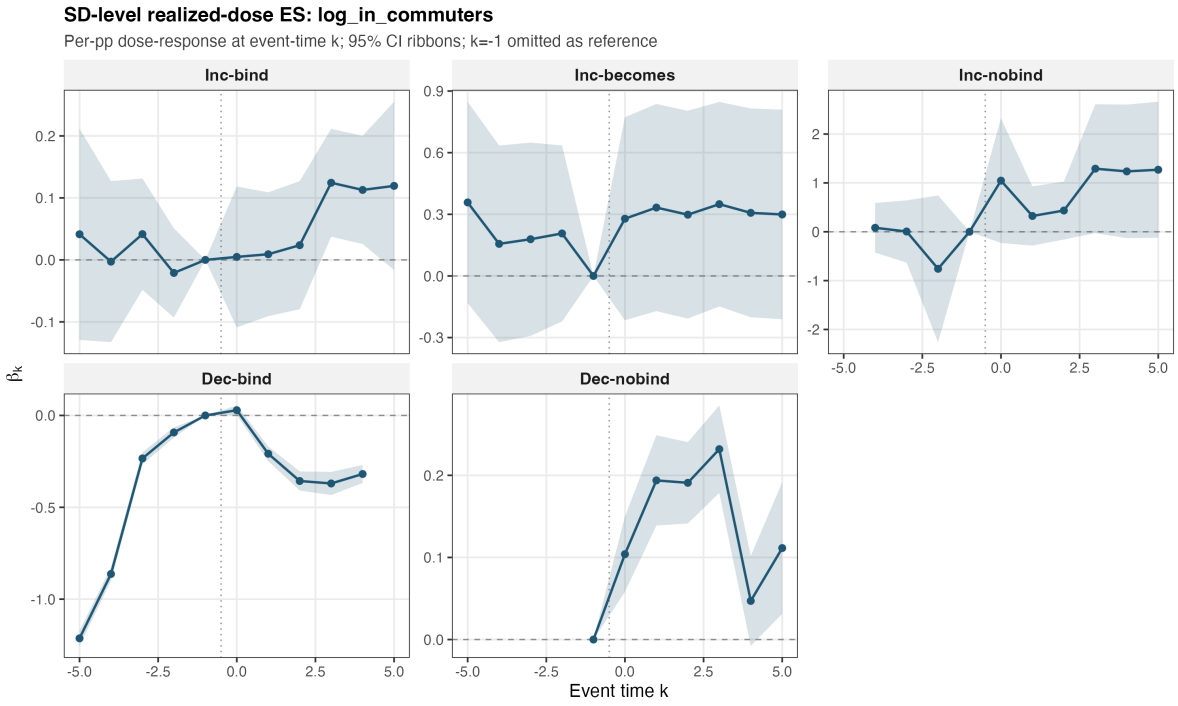


Figure 24: SD-level realized-dose ES on log(total residents), by cell.

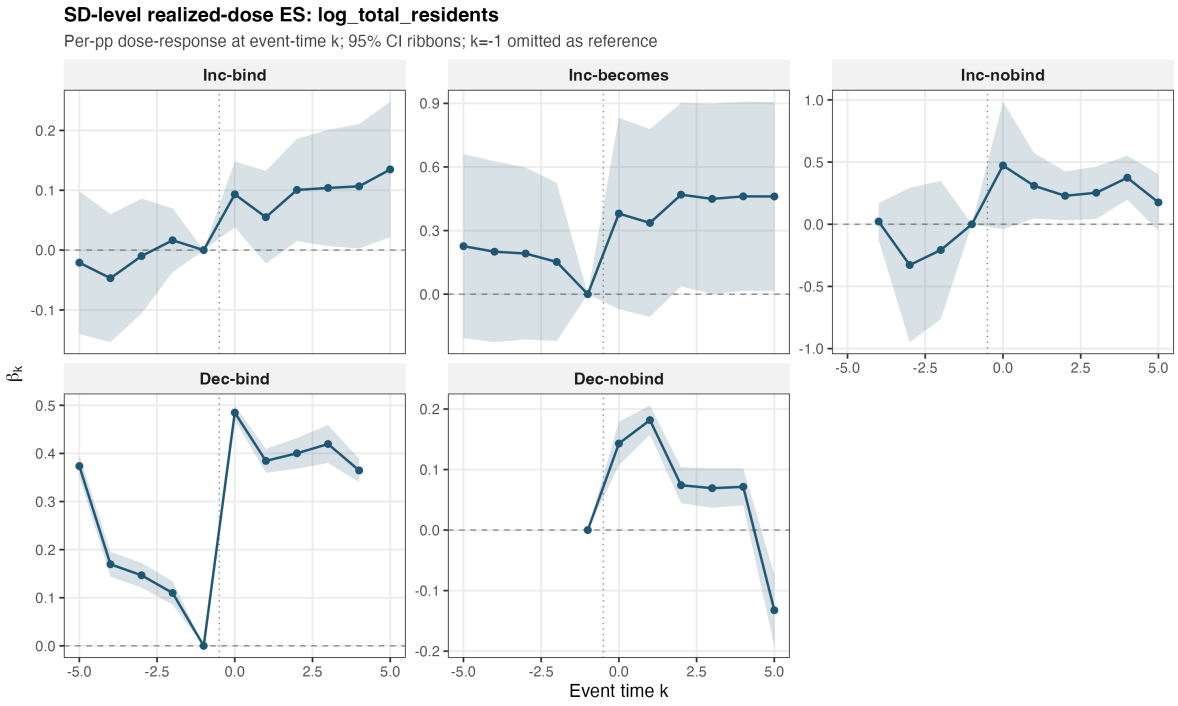
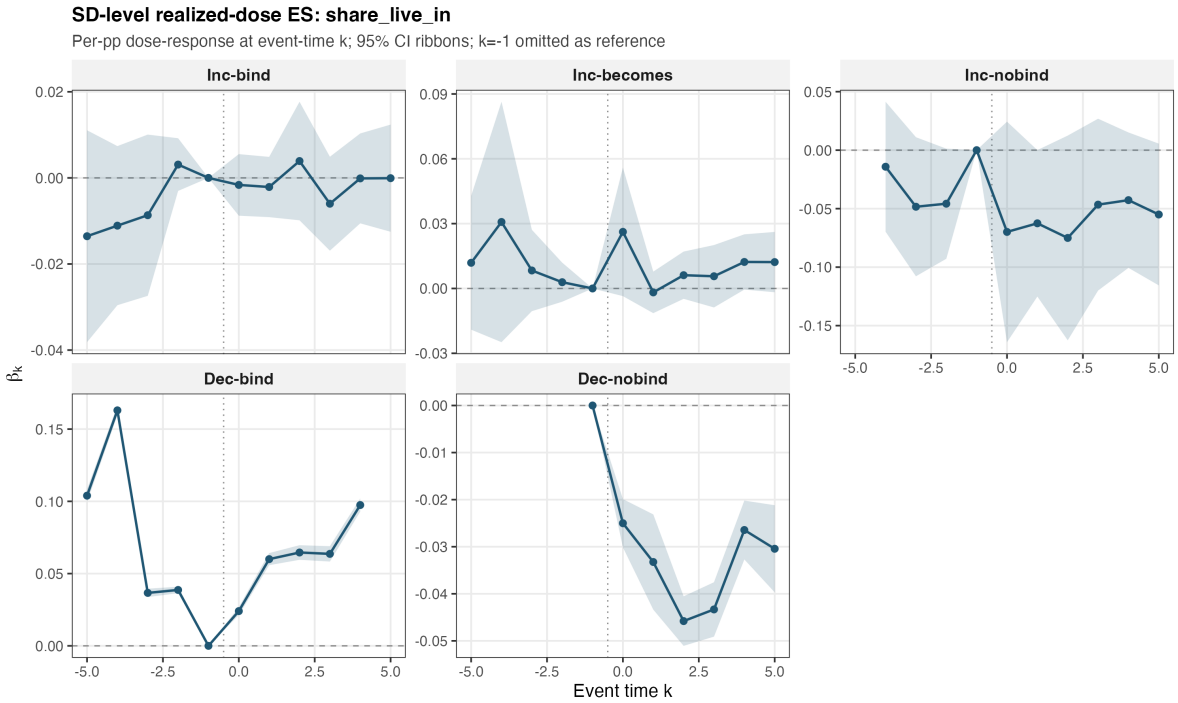



Figure 25: SD-level realized-dose ES on the share of workers living in their own municipality, by cell.



## B.3 Block-level, by cell

Figure 26: Block-level realized-dose ES on log(work count) (in-block jobs), faceted by binding-state cell; municipality-cohort (red) and SD-cohort (blue).

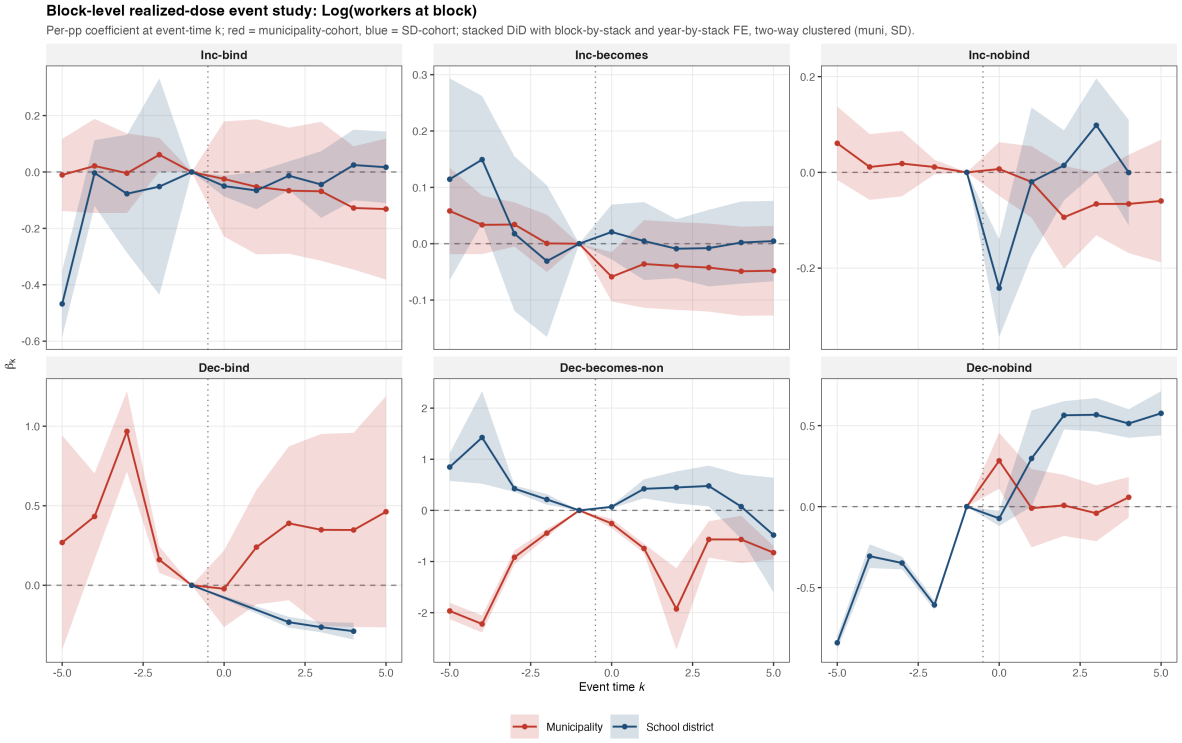
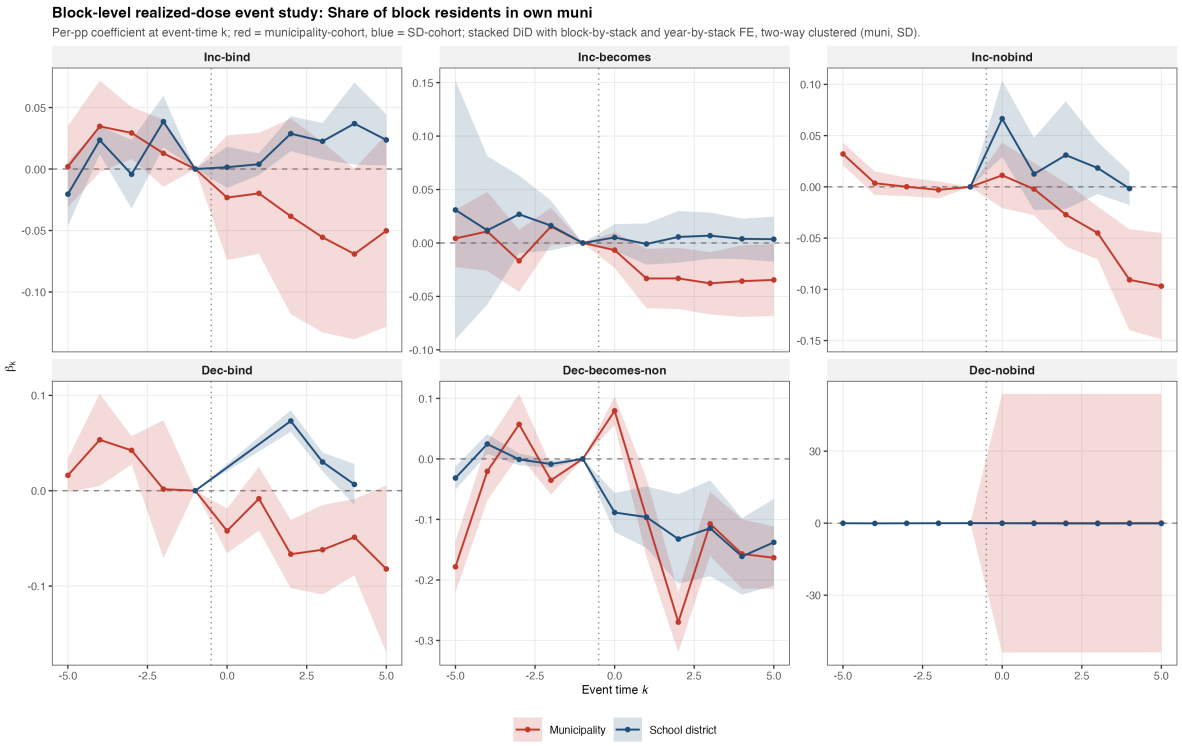


Figure 27: Block-level realized-dose ES on the share of block residents working in their own municipality, faceted by binding-state cell.




# C Robustness

Full tables behind the robustness summaries in Section **??**.

## C.1 Pre/post Act 32 split

Headline cell × dose specification re-estimated on pre-Act-32 ($c < 2012$) and post-Act-32 ($c \geq 2012$) cohort subsamples at each estimating unit.

Table 25: Pre-/post-Act-32 split — municipality level

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc-bind** | | | | |
| Pre-Act 32 | -0.339** | -0.156 | -0.098** | -0.047*** |
| | (0.138) | (0.112) | (0.045) | (0.013) |
| Post-Act 32 | -0.534*** | -0.334*** | -0.104*** | -0.040*** |
| | (0.075) | (0.080) | (0.034) | (0.004) |
| **Inc-becomes** | | | | |
| Pre-Act 32 | -0.230** | -0.239*** | -0.131* | -0.029* |
| | (0.093) | (0.089) | (0.074) | (0.016) |
| Post-Act 32 | -0.143 | -0.160 | -0.013 | 0.003 |
| | (0.106) | (0.108) | (0.019) | (0.009) |
| **Inc-nobind** | | | | |
| Pre-Act 32 | -0.051 | -0.078 | -0.052** | -0.013 |
| | (0.220) | (0.157) | (0.026) | (0.010) |
| Post-Act 32 | -0.094 | -0.305 | 0.365*** | -0.010 |
| | (0.113) | (0.368) | (0.095) | (0.013) |
| **Dec-bind** | | | | |
| Pre-Act 32 | 0.341*** | 0.049 | -0.013 | 0.074* |
| | (0.046) | (0.135) | (0.277) | (0.044) |
| Post-Act 32 | 0.068 | 0.056 | -0.133 | 0.008*** |
| | (0.144) | (0.183) | (0.102) | (0.003) |
| **Dec-becomes-non** | | | | |
| Pre-Act 32 | -1.220*** | 0.177** | -1.474*** | -0.144*** |
| | (0.069) | (0.071) | (0.032) | (0.005) |
| Post-Act 32 | 1.384*** | 0.874*** | 0.734*** | 0.057*** |
| | (0.106) | (0.112) | (0.053) | (0.008) |
| **Dec-nobind** | | | | |
| Pre-Act 32 | 0.377*** | -0.211 | 0.397*** | 0.019 |
| | (0.106) | (0.218) | (0.018) | (0.047) |
| Post-Act 32 | 0.094*** | 0.177*** | -0.132 | 0.006 |
| | (0.019) | (0.027) | (0.090) | (0.006) |
| *Pre-Act 32* | | | | |
| $N$ | 218,438 | 238,760 | 242,447 | 238,770 |
| $R^2$ | 0.964 | 0.959 | 0.981 | 0.823 |
| Within $R^2$ | 0.001 | 0.001 | 0.001 | 0.001 |
| Dep. mean | 3.675 | 5.845 | 6.696 | 0.093 |
| Dep. SD | 1.933 | 1.886 | 1.298 | 0.075 |

Table 26: Pre-/post-Act-32 split — school district level

| Outcome | Log(work in own muni) | Log(in-commuters) | Log(employed residents) | Share live in muni |
|---|---|---|---|---|
| **Inc-bind** | | | | |
| Pre-Act 32 | 0.027 | 0.057 | 0.106*** | 0.003 |
| | (0.038) | (0.046) | (0.030) | (0.004) |
| **Inc-becomes** | | | | |
| Pre-Act 32 | 0.290* | 0.155 | 0.272** | 0.008 |
| | (0.162) | (0.144) | (0.128) | (0.006) |
| Post-Act 32 | 0.043 | 0.523*** | 0.866*** | -0.052 |
| | (0.607) | (0.112) | (0.278) | (0.054) |
| **Inc-nobind** | | | | |
| Pre-Act 32 | 0.794 | 1.149* | 0.421*** | -0.035 |
| | (0.506) | (0.633) | (0.148) | (0.023) |
| **Dec-bind** | | | | |
| Post-Act 32 | 0.056 | 0.049 | 0.178*** | -0.000 |
| | (0.075) | (0.111) | (0.044) | (0.004) |
| **Dec-nobind** | | | | |
| Pre-Act 32 | -0.126*** | 0.155*** | 0.076*** | -0.032*** |
| | (0.025) | (0.028) | (0.016) | (0.002) |
| *Pre-Act 32* | | | | |
| $N$ | 51,382 | 51,416 | 51,416 | 51,416 |
| $R^2$ | 0.986 | 0.987 | 0.987 | 0.953 |
| Within $R^2$ | 0.004 | 0.003 | 0.004 | 0.004 |
| Dep. mean | 6.261 | 8.335 | 8.769 | 0.125 |
| Dep. SD | 1.298 | 1.202 | 0.909 | 0.068 |
| *Post-Act 32* | | | | |
| $N$ | 50,947 | 50,981 | 50,981 | 50,981 |
| $R^2$ | 0.986 | 0.987 | 0.987 | 0.953 |
| Within $R^2$ | 0.003 | 0.001 | 0.003 | 0.004 |
| Dep. mean | 6.262 | 8.335 | 8.770 | 0.125 |
| Dep. SD | 1.297 | 1.203 | 0.908 | 0.068 |

*Notes:* Cluster-robust standard errors in parentheses. Treated cohorts split at first event year $c < 2012$ (Pre-Act 32) vs $c \geq 2012$ (Post-Act 32). Cells omitted (no estimable coefficients): Dec-becomes-non. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 27: Pre-/post-Act-32 split — block level

| Outcome | Log(workers live) | Log(workers work) | Share live+work in muni |
|---|---|---|---|
| **Inc-bind** | | | |
| Pre-Act 32 | -0.218*** | -0.118** | -0.037 |
| | (0.059) | (0.058) | (0.037) |
| Post-Act 32 | -0.116** | -0.022 | -0.035*** |
| | (0.051) | (0.126) | (0.012) |
| **Inc-becomes** | | | |
| Pre-Act 32 | -0.001 | -0.041 | -0.034*** |
| | (0.058) | (0.033) | (0.013) |
| Post-Act 32 | -0.015 | -0.062** | -0.053*** |
| | (0.026) | (0.028) | (0.014) |
| **Inc-nobind** | | | |
| Pre-Act 32 | -0.114*** | -0.132** | -0.033*** |
| | (0.033) | (0.066) | (0.008) |
| Post-Act 32 | 0.672*** | 0.106** | -0.147*** |
| | (0.086) | (0.052) | (0.036) |
| **Dec-bind** | | | |
| Pre-Act 32 | 0.368** | 0.662*** | -0.024 |
| | (0.171) | (0.095) | (0.039) |
| Post-Act 32 | 0.045 | -0.441*** | -0.077*** |
| | (0.093) | (0.130) | (0.018) |
| **Dec-becomes-non** | | | |
| Pre-Act 32 | -1.004*** | 0.804*** | -0.151*** |
| | (0.155) | (0.185) | (0.042) |
| Post-Act 32 | 0.494*** | 0.364*** | -0.073*** |
| | (0.067) | (0.065) | (0.028) |
| **Dec-nobind** | | | |
| Pre-Act 32 | 0.341*** | -0.585** | 0.006 |
| | (0.030) | (0.260) | (0.008) |
| Post-Act 32 | -0.025 | 0.235*** | 0.006 |
| | (0.054) | (0.011) | (0.006) |

## C.2 Sample variants

Three sample variants for the DCED nonresident-rate panel (strict, filled, drop-incomplete) leave the headline municipality Inc-bind coefficient on log live-and-work essentially unchanged at $-0.401^{***}$.